

Socially Fair and Hierarchical Facility Location Problems

Abstract

The classic facility location problem seeks to open a set of facilities to minimize the cost of opening the chosen facilities and the total cost of connecting all the clients to their nearby open facilities. Such an objective may induce an unequal cost over certain groups of clients (i.e., total distance traveled by clients in a group). This is important when planning the location of socially relevant facilities such as emergency rooms. In this work, we consider a *fair* version of the problem by minimizing the Minkowski p -norm of the distances traveled and the cost of opening facilities, to penalize high access costs to open facilities across r groups of clients. This generalizes classic facility location ($p = 1$) and the minimization of the maximum total distance traveled by clients in any group ($p = \infty$). We show that there is a small portfolio of solutions where for any norm, at least one of the solutions is a constant-factor approximation with respect to any p -norm. We also give a lower bound on the cardinality of such portfolios. We also introduce the notion of *weak* and *strong* refinements for the facility location problem, that imposes further structural properties across solutions. The former requires that the set of facilities open for a lower p -norm is a superset of those open for higher p -norms, and the latter further imposes a partition refinement over the assignment of clients to open facilities in different norms. We give an $O(1)$ -approximation for weak refinements, $\text{poly}(r^{1/\sqrt{\log r}})$ -approximation for strong refinement in general metrics and $O(\log r)$ -approximation for the tree metric, where r is the number of (disjoint) client groups.

Our techniques also generalize to hierarchical versions of the problem. Here we are given multiple instances of facility location problems on the same network that differ only in facility opening costs that decrease in the hierarchy. The goal is to obtain a family of solutions, one for each instance, where an opened facility in a solution remains open in instances with lower cost. We give a $\text{poly}(e^{\sqrt{l}})$ -approximation for the problem with arbitrary metrics where l is the number of instances in the hierarchy. We also give $O(l^2)$ -approximation for the problem on line and tree metrics.

1 Introduction

In recent decades, optimization and algorithm design have revolutionized numerous industrial sectors – ranging from supply chains and network design to food production and finance. Novel optimization techniques have led to many industries achieving previously inconceivable levels of efficiency. The race to efficiency, however, has led to an inequitable distribution of resources among different segments of the population. For instance, the goal of profit maximization by grocery stores has led to the formation of food deserts in poorer parts of the United States [15]. Furthermore, a continuation of this trend can potentially lead to the formation of medical deserts [40]. It is clear that the pursuit of maximizing a singular objective can lead to crucial social and ethical problems, and therefore, it is imperative that optimization research studies a wider range of objectives that are important from a fairness viewpoint. In recent years, fairness has been extensively studied from an optimization and algorithm design perspective [6, 32, 34, 22], in both theory and application.

There are two broad ways to make constrained optimization problems fairer: one is to incorporate equity or fairness metrics as the optimization objective, and the other is to constrain solutions so they satisfy a minimum desired level of fairness. Further, there are broadly two different notions of fairness in optimization – individual and group fairness [17, 16]. In this work, we focus on various fairness objectives under the same set of constraints for the (uncapacitated) facility location problem. We are concerned with group fairness, wherein each member of the population belongs to a single group.

The uncapacitated facility location problem is one of the most well-studied combinatorial optimization problems. In the classic version of the problem, we have a set of demands or clients D , a set of facilities F , and distances d_{ij} between each facility $i \in F$ and client $j \in D$ that satisfy the triangle inequality. Opening facility $i \in F$ incurs a cost c_i . The objective is to open a set of facilities and assign each client to some open facility such that the total cost of opening the facilities and the distances of clients from their assigned facilities is minimized. The facility location problem models various network design situations such as opening offices, warehouses, hospitals, proxy servers, transport hubs etc. The simple and elegant facility location problem has received significant attention in theory and practice [28, 13, 41, 10, 11, 35].

A solution to the uncapacitated facility location problem can be disproportionately bad for a certain set of clients while being optimal with respect to the overall cost. As a simple example, consider clients on a line where a facility of cost $m \in \mathbb{Z}_+$ can be opened anywhere on the line, and there are m clients at 0 and $m - 1$ clients at 1; the optimal solution to the classic facility location opens a single facility at 0. This disproportionately affects the set of clients at 1.

As discussed, this is particularly undesirable in social applications where underprivileged groups of clients can be disadvantaged by the optimal solution. As a further example, a recent study on emergency rooms in Alameda County showed that the closure of the Alta Bates Emergency Room would disproportionately displace the uninsured population (clients in facility location terminology) and people of color [12].

To model such phenomenon, we study the *socially fair* version of the facility location problem, where we are given several *client groups* that partition the client set, and the total cost of a solution is a function of the total distances of each group. Formally, the socially fair facility location problem is defined as the following:

Definition (Fair facility location). We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty set of facilities $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}_+$. Furthermore, suppose we are given non-empty client groups $D_1, \dots, D_r$ that partition D . For $p \in [1, \infty]$, the

p -norm fair facility location problem requires opening a subset of facilities $F' \subseteq F$ and providing a client to facility assignment $\Pi : D \rightarrow F$ so as to minimize the cost of open facilities and the Minkowski p -norm of average group distances, i.e.,

$$\min \sum_{i \in F'} c_i + \left[\sum_{s \in [r]} \left(\frac{1}{|D_s|} \sum_{j \in D_s} d(j, \Pi(j)) \right)^p \right]^{1/p}.$$

For $p = 1$ and singleton groups (i.e., $|D_s| = 1$ for all $s \in [r]$), the fair facility location problem reduces to the aforementioned classic facility location problem. For $p = \infty$, the objective corresponds to minimizing the maximum distance across all groups, weighted by their group size. This variant has been studied in the context of emergency room and trash dumpsite placement [12, 14].

We remark that while we have defined our objective for *average* distance to open facilities for each group, our results hold for a much larger class of objectives, including total distance to open facilities for each group.

Our primary goal is not restricted to getting a constant-factor approximation for the p -norm fair facility location problem: indeed, that can be achieved by a relatively straightforward generalization of Shmoys *et al.*'s algorithm [41] for the classic facility location problem. Instead, we also aim to understand how approximate solutions change as p varies, and if common structure can be preserved between these approximate solutions for various norms. As we shall see, this is useful from a practical policy design perspective, and interesting from a theoretical point of view. This is also an interesting question from a multi-criteria optimization viewpoint: our problem is an instance of the broader question about what common structures do optimal (or approximate) solutions maintain for related objectives.

Contributions. As a warm-up to the fair facility location, we show that the 4-approximation algorithm of Shmoys *et al.* [41] for the classic facility location problem generalizes to a large class of convex objective functions, and in particular gives a 4-approximation to the p -norm fair facility location problem for each $p \in [1, \infty]$; see Section 1.3 and Appendix A¹. This observation is useful in showing our key contributions, explained next.

1. **Portfolio of Solutions.** A challenge in the fair facility location problem is deciding what p -norm to choose to model a fair objective. The chosen norm allows to weigh the connection cost paid by different social groups and appropriate weighing may not be clear even to the algorithm designer. Indeed, for classification algorithms, it has been shown that different fairness objectives may not be simultaneously satisfied [32]. From a policy design perspective, then, we propose to leave the task of choosing the correct fairness metric to the policy makers instead of incorporating it as a fixed objective in the algorithm.

Therefore, given an instance of the fair facility location problem, a natural question is the following: *is an optimal solution to the p -norm fair facility problem a good approximation to the q -norm fair facility problem for different $p, q \in [1, \infty]$?* Figure 1 shows that this is not always the case; indeed, the cost of an optimal solution to the p -norm problem can be very high for the q -norm problem. This raises the following question: *is there a small set of candidate solutions such that no matter what norm is chosen, one of the solutions in the set is a good*

¹We remark that several other improved algorithms for the classic facility location problem do not generalize for more general convex objectives because they crucially use the linearity of the objective.

	Arbitrary metric	Line metric	Tree metric ³
weak refinement	16		
strong refinement	$\text{poly}(r^{\frac{1}{\sqrt{\log r}}})$	$O(\log r)$	$O(\log r)$

Table 1: A summary of approximation guarantees for refinements for the socially fair facility location problem in Section 3. Here r is the number of client groups in the problem instance.

approximation corresponding to that norm. Empirical studies for the facility location problem suggest that this might possibly be true even for an infinite class of objective functions [27].

Building on ideas similar to those in [25], we show in Section 2, that when there are r client groups, there exist (approximately) $\log_2 r$ solutions such that for each $p \in [1, \infty]$, one of these solutions is an 8-approximation for the p -norm fair problem (Theorem 2). This is important in context of policy design: the task of deciding what is ‘fair’ can be left to policy makers along with a small set of candidate solutions that are a good approximation irrespective of what definition of fair is chosen. Conversely, we show that there exist problem instances where at least $\tilde{\Omega}(\sqrt{\log_2 r})$ ² different solutions are needed such that one of them is an $O(1)$ -approximation for any p -norm (Theorem 3).

2. **Refinements.** Additionally, we introduce the notion of *refinements* which establish further structure between approximate solutions across different p -norms, $p \in [1, \infty]$.

Given an instance of the facility location problem, for weak refinement, we demand that for any two norms $p < q$, we must have $F_q \subseteq F_p$ where F_p, F_q are the set of facilities in the solution for p, q -norm fair problems respectively. This also establishes a decreasing subset structure for open facilities across solutions for p -norms as p varies from 1 to ∞ . Therefore, if the fairness criterion changes at some point, a completely different set of facilities does not have to be open or closed.

For strong refinement, we additionally demand that the set of assigned clients to any facility in F_p must be assigned to a single open facility in F_q ($p < q$). Thus the partition of client set obtained by their assignment to open facilities in solution for norm q must be a refinement of the partition obtained in the solution for norm p . From a planning perspective, strong refinements induce a nice hierarchical structure over the potential set of solutions which can eventually help policy makers in growing the number of facilities that serve a set of customers, as the need arises.

We formalize this notion in 1.1; Section 3 deals with approximation algorithms for refinements. For arbitrary metrics, we give an $O(1)$ -approximation for weak refinement (Theorem 4) and a $\text{poly}\left(r^{\frac{1}{\sqrt{\log r}}}\right)$ -approximation for strong refinement (Section 3.1, Corollary 2). For the special cases of line and tree metrics, we improve the approximation factor to $O(\log r)$ for strong refinement (Section 3.2, Corollary 3 and Section 3.3, Corollary 4 respectively). A summary of these results is presented in Table 1.

3. **Trade-off Between Facility Opening Costs and Connection Costs:** Additionally, we characterize the trade-off between the facility and assignment costs across different norms by considering a weighted objective in Section 4. We show that if the weight is lower than a threshold, the optimal objective value is monotone decreasing as a function of p norm. We show similarly that this behavior is completely reversed if the weight parameter is higher than

²We say $f(r) = \tilde{\Omega}(\sqrt{\log_2 r})$ if and only if $\sqrt{\log_2 r} = O(f(r) \cdot \text{poly}(\log \log r))$, i.e., we suppress $\log \log r$ terms.

a norm-dependent threshold. We also show that this monotonicity does not always hold in all other cases; see Theorem 8.

We propose an alternate model for the fair facility location problem with similar approximation guarantees for the number of solutions and refinements (Corollary 5, Theorems 9, 10). We also consider the related socially fair p -norm k -clustering problem studied in [23, 24, 25].

4. **Hierarchical Facilities.** Inspired by the notion of refinements, we introduce a version of the *hierarchical facility location* problem. In several applications of the facility location problem such as the placement of corporate offices, judicial courts, hospitals, public works offices, facilities need to be opened at several hierarchical levels with additional constraints across those levels [18, 47]. For instance, public works offices usually operate at several levels, with the office at the lowest level being the first point of contact for an applicant (i.e., a client in facility location terms). The hierarchy may consist of local area-wide offices at the lowest level, city-wide offices at the next level, and state-wide offices at the next level, and a country-wide office at the final level. An unsatisfactory resolution by a lower office can prompt a client to approach an office at the next higher level that the lower office reports to.

In most such applications, the cost of opening a facility at a higher level is higher than the cost of opening a facility at a lower level. Furthermore, facilities at a given level usually report to a unique facility at the next level. Therefore, once a client chooses a facility at the lowest level, their choice of facility at all higher levels is predetermined.

For l hierarchical levels, our version of the hierarchical facility location problem demands solutions for each of the l levels of the given instance of the problem that satisfy the above hierarchic constraints, while also being a good approximation for the independent facility location problems at their respective levels. We formalize this definition in Section 1.1.

Hierarchical facility location and refinements for the fair facility location problem share very similar structures. In Section 5, we show that our techniques for refinements also lead to results for hierarchical facility location, giving an $\tilde{O}(e^{3\sqrt{l}})$ -approximation for general metrics, and an $O(l^2)$ -approximation for tree metrics (Theorem 11).

We define the various problems formally in Section 1.1. We give a brief literature survey for classic, fair, and hierarchical facility location problems in Section 1.2. Finally, we include the generalization of Shmoys *et al.*'s algorithm [41] as a preliminary in Section 1.3 before proceeding to the rest of our paper.

1.1 Problem definitions and notation

We define a general version of the facility location problem, where the objective is defined by a suitable convex function. Given a convex, non-decreasing function $f : \mathbb{R}_+^D \rightarrow \mathbb{R}_+$, we minimize the sum of facility costs and f applied to individual client group distances. The classic facility location corresponds to the case where $f(x) = \sum_j x_j$ for $x \in \mathbb{R}_+^D$, i.e., the total assignment cost is the sum of distances of individual clients. Furthermore, the p -norm facility location problem is a special case of the generalized facility location problem. We present the formal definition:

Definition (Generalized facility location). We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty facility set $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}$. Furthermore, we are given a non-decreasing convex function $f : \mathbb{R}_+^D \rightarrow \mathbb{R}_+$. The general facility location problem

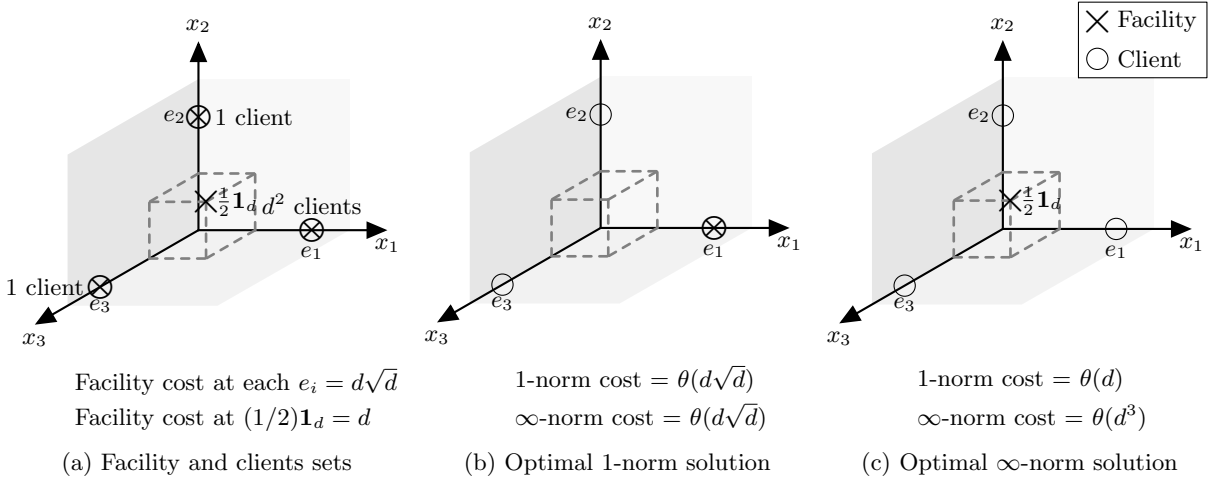


Figure 1: An example to illustrate how optimal solution for one norm can be a bad approximation for another norm. The underlying metric space is $\mathbb{R}^d$ with Manhattan distance metric (i.e. 1-norm). The client set is $D = \{e_i : i \in D\}$ where e_i is the i th unit vector. There are d^2 clients at e_1 , and one client each at $e_2, \dots, e_d$. Each client forms its own group. The set of facilities is $D \cup \{\frac{1}{2}\mathbf{1}_d\}$, where $\mathbf{1}_d$ is the all-ones vector. The cost of opening a facility at each e_i is $d\sqrt{d}$ while the cost of opening a facility at $\frac{1}{2}\mathbf{1}_d$ is d . Although in Figure 1a, we depict client and facility sets for $d = 3$, we are really interested in large d . For large d , the optimal 1-norm fair solution is to open a single facility at e_1 (see Figure 1b), and it has total cost $2(d-1) + d\sqrt{d}$. The optimal ∞ -norm fair solution is to open a single facility at $\frac{1}{2}\mathbf{1}_d$ (see Figure 1c) with cost $d + \frac{d}{2} = \frac{3d}{2}$. The optimal 1-norm solution is a bad approximation for the ∞ -norm fair problem: it has cost $d\sqrt{d} + 2$ for the ∞ -norm fair problem. Conversely, the optimal ∞ -norm solution is a bad approximation for the 1-norm fair problem: it has cost $d^2 \cdot \frac{d}{2} + (d-1) \cdot \frac{d}{2} + d = \theta(d^3)$ for the 1-norm problem.

requires opening a subset of facilities $F' \subseteq F$ and providing an assignment $\Pi : D \rightarrow F'$ so as to minimize

$$\underbrace{\sum_{i \in F'} c_i}_{\text{Facility cost } C_{F'}} + \underbrace{f\left(\left(d(j, \Pi(j))\right)_{j \in D}\right)}_{\text{Assignment cost } C_{\Pi}},$$

where $\left(d(j, \Pi(j))\right)_{j \in D}$ is the vector in $\mathbb{R}^D$ consisting of individual client distances.

For any instance of the facility location problem, a solution $\mathcal{S}$ is an ordered pair (F', Π) where $F' \subseteq F$ is the set of open facilities and $\Pi : D \rightarrow F'$ is the assignment function. The *total cost* of the solution, $C_{\mathcal{S}}$ is the sum of the *facility cost* $C_{F'}$ and the *assignment cost* C_{Π} , that is, $C_{\mathcal{S}} = C_{F'} + C_{\Pi}$. If f is non-decreasing, then each client is assigned to its nearest open facility for any optimal solution. Therefore, a solution is sometimes identified just by the set of open facilities in facility location literature. However, we will deal extensively with different assignments and always explicitly give the assignment function.

We discuss two different notions of refinements for the p -norm fair facility location problem. *Weak (or facility) refinements* preserve the structure for open facility sets across p , by ensuring that any facility that was open for a higher p -norm remains open for lower p -norms as well. *Strong (or partition) refinements* are weak refinements that also preserve structure for the corresponding assignments, by ensuring that any customer that was mapped to a certain facility in a lower p -norm, must be mapped to the same facility (if it is open) in the higher p -norm. We can formalize these notions as follows:

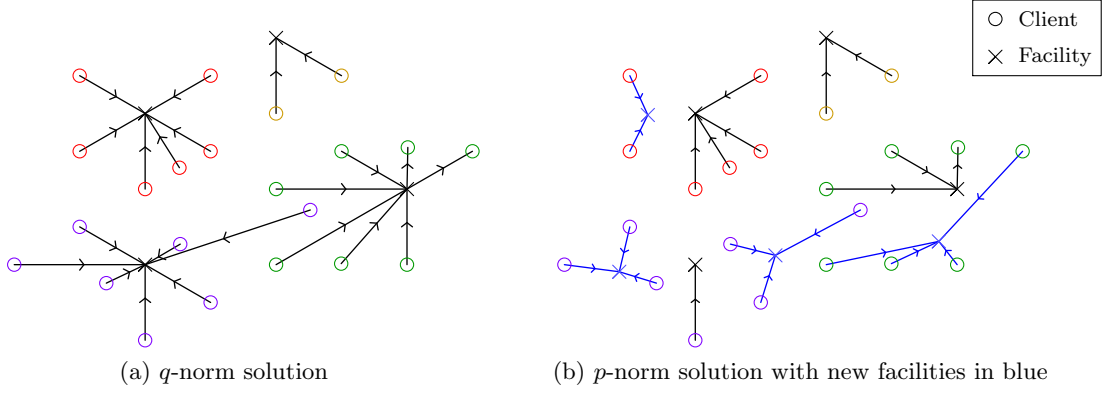


Figure 2: A toy example to illustrate refinements. Consider $P = \{p, q\}$ with $q > p$. (a) A solution for the q -norm problem with open facilities marked as crosses and the assignment function indicated by arrows; clients assigned to the same facility have the same color for clarity. (b) A solution for the p -norm problem: this solution refines the q -norm solution, with new facilities and new assignment arrows in blue. Note that no facility has been deleted from the q -norm solution, and all clients assigned to a new facility are assigned to the same facility in the q -norm solution (i.e., all clients assigned to the same facility still have the same color).

Definition (Refinements). We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty facility set $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}$. We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty facility set $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}$. Furthermore, suppose we are given non-empty client groups $D_1, \dots, D_r$ that partition D . Given a set of norms $P \subseteq [1, \infty]$, a solution set $S = \{\mathcal{S}_p = (F_p, \Pi_p) : p \in P\}$ is called

1. a *weak refinement* if for all $p, q \in P$ with $p \leq q$, $F_p \supseteq F_q$; that is, the facilities form a subset structure.
2. a *strong refinement* if for all $p, q \in P$ with $p \leq q$ and for each facility $i \in F_p$, there is some facility $i' \in F_q$ such that all clients assigned to i under Π_p are assigned to i' under Π_q , that is,

$$\{j \in D : \Pi_p(j) = i\} \subseteq \{j \in D : \Pi_q(j) = i'\}.$$

If for each norm $p \in P$, the solution $\mathcal{S}_p$ is an α -approximation for the p -norm fair problem, then we say that the refinement is an α -approximate refinement.

One can think of refinements as dynamic solutions to an instance of the fair facility location problem that change with p . As $p \in P$ decreases, weak refinements *open* more facilities but do not close any open facilities. For strong refinements, this can be interpreted in terms of the partition of D that Π_p induces. As p decreases and new facilities are potentially added, this partition of D is refined. See Figure 2 for a toy example. As described earlier, refinements demand some common structure across different p -norms (i.e., across different fairness criteria).

Finally, we discuss the hierarchical facility location problem. Recall the assumptions: we wish to solve several facility location problems at different levels in a hierarchy. The cost of opening a facility at a level increasing with the level. Furthermore, each open facility at a given level is also open at a lower level. Finally, all clients of a facility at a given level must be clients of the same facility at a higher level - this condition is often true for most hierarchical structures such as corporate offices and speciality hospitals. We present the formal definition.

Definition (Hierarchical facility location). We are given a metric space $\mathcal{M} = (X, d)$, and non-empty facility set $F \subseteq X$ and demand set $D \subseteq X$, and l different general facility location problems for some positive integer l , one for each *level* in $[l]$. The l problems are specified by uniform facility opening costs $c_1, \dots, c_l \in \mathbb{R}_+$ satisfying $c_1 \leq \dots \leq c_l$; and a non-negative, non-decreasing, convex function $f : \mathbb{R}_+^D \rightarrow \mathbb{R}$. In the hierarchical facility location problem, we are asked to provide solutions $\{\mathcal{S}_k = (F_k, \Pi_k) : k \in [l]\}$ such that for all $k, t \in [1, l]$ with $t \geq k$:

1. $F_k \supseteq F_t$, and
2. for all $i \in F_k$, there exists $i' \in F_t$ such that

$$\{j \in D : \Pi_p(j) = i\} \subseteq \{j \in D : \Pi_q(j) = i'\}.$$

The objective is to minimize the worst approximation ratio among l levels, that is,

$$\max_{k \in [l]} \frac{C_{\mathcal{S}_k}}{OPT_k},$$

where OPT_k is the optimal solution value for the generalized facility location problem with facility costs c_k .

We call this an l -level hierarchical facility location problem.

It should be clear that the hierarchical facility location is similar to the notion of refinements for the fair facility location problem: the constraints on open facility sets and assignments are similar. This is precisely why our algorithms work for both refinements and hierarchical facility location.

Throughout, we denote $n = |F \cup D|$. We assume that the input always has size at least n : the points in F and D always need to be explicitly specified; the distance function d can be specified implicitly or explicitly. For instance, when the metric space $\mathcal{M}$ is the Euclidean space $\mathbb{R}^2$, the points in F and D need to be specified as 2-tuples of real numbers while the distance d is implicit. Therefore, all algorithms that run in polynomial time in n are polynomial in the input size.

We usually denote facilities using the letter i and occasionally using the letters f, g, h . Clients are usually denoted by j . The letters k, l usually denote indices for different p -norms or levels in the hierarchical facility location problem; occasionally letters s, t are also used for this. Solutions (i.e., pairs of facilities and assignments) are usually denoted by letters $\mathcal{S}, \mathcal{R}$. Subsets of the facility set F are denoted by letters F, G . Assignments of clients to facilities (or facilities to facilities at different levels in a hierarchical problem) are denoted by letters Π and θ .

1.2 Literature review

We present a brief review of the literature for classic, fair, and hierarchical facility location problems.

The classic facility location problem has been extensively studied. An $O(\log n)$ -approximation algorithm was given by Hochbaum [28]; notably, this algorithm does not assume the metric property. The first constant-factor approximation was given by Shmoys *et al.* [41] who gave a deterministic 4-approximation algorithm. They also give a randomized 3.16-approximation algorithm. Since then, a steady stream of algorithms has led to frequent improvements in this approximation factor [33, 10, 11, 29, 37, 9]. The state-of-the-art is a 1.488-approximation algorithm of Li [35]; while the problem is inapproximable to within a factor 1.463 unless $P = NP$ [26, 42]. The textbook by

Shmoys and Williamson [45] gives an overview of and more details about many of the techniques used.

Several fairness objectives for the facility location problem have been studied [30, 20, 43]; for an extensive list see the survey articles [39, 5]. To the best of our knowledge, none of these fairness objectives have been studied from an approximation algorithm viewpoint for the uncapacitated facility location problem. However, several approximation algorithms are known for the fair versions of the related clustering problem where a fixed number of facilities are allowed to open [30, 22, 2, 38, 23, 25, 24].

Minkowski p -norm objectives have been widely considered in combinatorial optimization literature as a model for fairness and as interesting theoretical questions. k -clustering [22, 23, 24, 25], traveling salesman problem [19], set cover, scheduling and other problems [25] have all been studied with the p -norm objective.

There are a very large number of models for hierarchical facility location problems; see survey articles [47, 18]. Many of these models have not been studied from an approximation algorithm viewpoint. However, there are several models with an approximation guarantee for the overall cost [1, 8, 3, 4, 46, 31, 21, 44]. Almost all of these models aim to minimize some notion of the total cost or maximize some notion of client coverage that is linear in open facilities and assignments. To the best of our knowledge, the model with a facility location problem at each level with minimization of the worst approximation ratio across levels has not been proposed so far.

1.3 Preliminaries

We prove the following theorem, which asserts that the generalized facility location problem can be 4-approximated for a large class of functions f . Specifically, we show that Shmoys *et al.*'s algorithm [41] – which uses the filtering technique introduced by Lin and Vitter [36] – extends to this class of functions. We include the algorithm and the proof of the theorem in Appendix A.

Before we state the theorem, we recall the definition of *sublinear* functions: f is sublinear if for all x, y in its domain (1) $f(cx) = cf(x)$ for all $c \geq 0$ and (2) $f(x + y) \leq f(x) + f(y)$. All norms are sublinear and all sublinear functions are convex. We are now ready to state the theorem:

Theorem 1. *If $f : \mathbb{R}_+^D \rightarrow \mathbb{R}$ is a differentiable sublinear function that is non-negative and non-decreasing, then there is 4-approximation to the generalized facility location problem with a polynomial number of oracle calls to $f, \nabla f$.*

Since all p -norm functions are sublinear, the theorem immediately implies the following corollary:

Corollary 1. *There is a polynomial-time algorithm that gives a 4-approximation for the p -norm fair facility location problem for any $p \in [1, \infty]$.*

We remark that several other algorithmic techniques that achieve a better approximation ratios for the classic facility location problem do not generalize for more general convex objectives because they (implicitly or explicitly) use certain special properties of linear functions: linearity of expectation for various randomized rounding algorithms, existence of linear dual for primal-dual algorithms, concavity of the objective for local search algorithms etc.

2 Portfolio of solutions for fair facility location

The main results in this section are Theorem 2 and Theorem 3. Given an instance of the fair facility location problem, an approximate solution to $p = 1$ can look very different from an approximate solution to $p = \infty$. It is natural to ask for the smallest number of solutions such that no matter what $p \in [1, \infty]$ is chosen, one of the solutions is a constant-factor approximation to the p -norm fair facility location problem. The answer is trivial for $r = 1$ since all p -norm fair objectives are the same in this case; we restrict our problem to the non-trivial case of $r \geq 2$ from here on. For $r \geq 2$, we show that the answer is no more than $\lceil \log_2 r \rceil$, and that it is at least $\tilde{\Omega}(\sqrt{\log_2 r})$. One way to interpret this statement is that while approximate solutions to the p -norm fair problem change as p varies from 1 to ∞ , we can force the number of changes to be at most $\lceil \log_2 r \rceil - 1$. We formalize the definition of such solution sets:

Definition (Approximate solution sets). Given an instance of the fair facility location problem and an approximation factor $\alpha \geq 1$, an α -approximate solution set is a set S of solutions such that for all norms $p \in [1, \infty]$, there is some solution in S that is an α -approximation to the p -norm fair facility location problem.

As previously alluded to, this is also interesting from a policy design perspective. The task of deciding what choice of $p \in [1, \infty]$ can be left to policy makers, and this solution set can be preemptively provided for analysis and policy implementation.

2.1 Upper bound

The key idea is to use the relationship between different p -norms in $\mathbb{R}^r$. Before we prove the theorem, we state such a relationship (for completeness we include the proof in Appendix B).

Lemma 1. For $p, q \in [1, \infty]$ with $p \leq q$ and any vector $x \in \mathbb{R}^r$,

$$\|x\|_q \leq \|x\|_p \leq r^{\frac{1}{p} - \frac{1}{q}} \|x\|_q.$$

We remark that the above bound is sharp; for all p, q , there exist simple examples where either of the two inequalities hold.

The next lemma states that approximate solutions for one norm can be translated into approximate solutions for a different norm, incurring an additional factor in the approximation ratio.

Lemma 2. Given an instance of the fair facility location problem with r groups and $1 \leq p < q \leq \infty$, any α -approximate solution to the p -norm fair problem is an $(\alpha r^{\frac{1}{p} - \frac{1}{q}})$ -approximate solution to the q -norm fair problem, and conversely, any α -approximate solution to the q -norm fair problem is an $(\alpha r^{\frac{1}{q} - \frac{1}{p}})$ -approximate solution to the p -norm fair problem.

The key idea for the proof is to use Lemma 1. This allows us to bound the p -norm cost of an approximation solution to the q -norm problem, and vice-versa. This, along with the fact that $OPT_p \geq OPT_q$ implies the lemma. We include a proof in Appendix B. We remark that the crucial idea for the proof is present in the proof of Theorem 14 of [25].

We are ready to prove the main result for this section.

Theorem 2. Given any instance of the facility location problem, there exists an 8-approximate solution set of cardinality at most $\lceil \log_2 r \rceil$. This solution set can be obtained in polynomial time.

Proof. Recall that OPT_p is a decreasing function of p . Define sequence $p_0, \dots, p_t$ as follows: $p_0 = 1$, p_k is the unique norm satisfying $\frac{OPT_{p_0}}{OPT_{p_k}} = 2^k$, and $t = \left\lceil \log_2 \left(\frac{OPT_1}{OPT_\infty} \right) \right\rceil - 1$. Also define $p_{t+1} = \infty$. From Lemma 1, we get that $OPT_\infty \leq rOPT_1$, so that $t + 1 \leq \lceil \log_2 r \rceil$. Also notice that

$$\frac{OPT_{p_t}}{OPT_{p_{t+1}}} = \frac{OPT_1}{2^t OPT_\infty} \leq \frac{2}{\frac{OPT_1}{OPT_\infty}} \cdot \frac{OPT_1}{OPT_\infty} = 2.$$

Corollary 1 allows us to obtain 4-approximate solutions for any p -norm for the given instance of the problem. We will claim that a 4-approximate solution $\mathcal{S}_k = (F_k, \theta_k)$ to the p_k -norm fair problem is an 8-approximate solution to the p -norm problem for any $p \in [p_k, p_{k+1}]$. Since $\cup_{k=0}^t [p_k, p_{k+1}] = [1, \infty]$, this implies that $\{\mathcal{S}_0, \dots, \mathcal{S}_t\}$ is an 8-approximate solution set for the given instance of the facility location problem. Since $|\{\mathcal{S}_0, \dots, \mathcal{S}_t\}| = t + 1 = \lceil \log_2 r \rceil$, this implies the theorem.

We prove the claim now. Let $V_{\theta_k} \in \mathbb{R}^r$ denote the cost vector for assignment θ_k . Then, the cost of solution $\mathcal{S}_k$ for a norm $p \in [p_k, p_{k+1}]$ is

$$C_{F_k} + \|V_{\theta_k}\|_p \leq C_{F_k} + \|V_{\theta_k}\|_{p_k} \leq 4OPT_{p_k} \leq 8OPT_p.$$

The first inequality follows from Lemma 1, the second inequality follows since $\mathcal{S}_k$ is a 4-approximate solution to the p_k -norm fair problem, and the final inequality follows since $OPT_{p_k} \geq OPT_p \geq OPT_{p_{k+1}} \geq \frac{1}{2}OPT_{p_k}$. This proves the claim. $\square$

2.2 Lower bound

We now provide a lower bound on the number of solutions required.

Fact 1. $t^{\frac{1}{o(\log_2 t)}} = 2^{\frac{\log_2 t}{o(\log_2 t)}} = \omega(1)$ for large t .

Theorem 3. *For any large positive integer r , there is an instance of the fair facility location problem with r clients such that at any $O(1)$ -approximate solution set contains at least $\tilde{\Omega}(\log_2 r)$ distinct solutions.*

Proof. We consider r clients located at the central vertex of a star. We let $r = t^k$, where t and k are parameters that we will choose later; also let $\epsilon = \frac{1}{k}$. The star has k edges with leaves $f_1, \dots, f_k$, leaf f_j has facility with opening cost t^j and is at distance $t^{(k-j)\epsilon}$ from the central vertex, for each $j \in [k]$. Each client is in its own group. This is presented in Figure 3.

For $i \in [k]$, let $p_i = \frac{k}{i}$. For appropriate choices of t, k , we will show that for distinct i, i' , no $O(1)$ -approximate solution to the p_i -norm fair problem is an $O(1)$ -approximate solution to the $p_{i'}$ -norm fair problem. Therefore, any $O(1)$ -approximate solution set for this problem instance contains at least k distinct solutions.

Fix $i \in [k]$. We show that any $O(1)$ -approximate solution to the p_i -norm fair problem must have (a) the facility at f_i open and (b) none of the facilities at $f_{i+1}, \dots, f_k$ open. This is sufficient to prove our claim.

If only f_i is open, the solution cost for p_i -norm is

$$C_{p_i, f_i} = t^i + \|t^{(k-i)\epsilon} \mathbf{1}_r\|_{p_i} = t^i + r^{\frac{1}{p_i}} t^{(k-i)\epsilon} = t^i + t^{i+(k-i)\epsilon} = \Theta(t^{i+(k-i)\epsilon}). \quad (1)$$

Since $\epsilon = \frac{1}{k}$, this cost is of the order $\Theta(t^{i+\frac{k-1}{k}}) = \Theta(t^{(i+1)-\frac{1}{k}})$.

For a solution $\mathcal{S}$ with any of the facilities $f_{i+1}, \dots, f_k$ are open, the solution cost $C_{\mathcal{S}} = \Omega(t^{i+1})$, which is higher than C_{p_i, f_i} by a factor $\Omega(t^{\frac{1}{k}})$. For $k = o(\log t)$, Fact 1 implies that this factor is $\omega(1)$, and therefore $\mathcal{S}$ is not an $O(1)$ -approximate solution to the p_i -norm fair problem.

Further, consider a solution $\mathcal{S}$ with none of the facilities $f_i, \dots, f_k$ open. At least one of the facilities in $f_1, \dots, f_{i-1}$ must be open, and even if all clients are assigned to the nearest open facility, the distance of each client from the nearest open facility is at least $t^{(k-(i-1))\epsilon}$. Therefore, the solution cost

$$C_{\mathcal{S}} = \Omega\left(\|t^{(k-i+1)\epsilon} \mathbf{1}_r\|_{p_i}\right) = \Omega\left(\|t^{(k-i+1)\epsilon} \mathbf{1}_r\|_{p_i}\right) = \Omega\left(t^{i+(k-i+1)\epsilon}\right).$$

This is higher than the cost of only opening f_i , i.e., C_{p_i, f_i} by a factor of $\Omega\left(t^{\frac{1}{k}}\right)$. For $k = o(\log t)$, Fact 1 implies that this factor is $\omega(1)$, and therefore $\mathcal{S}$ is not an $O(1)$ -approximate solution to the p_i -norm fair problem.

Finally, we show that for $k = \sqrt{\frac{\log_2 r}{\log_2 \log_2 r}}$ and $k^t = r$, we have $k = o(\log t)$, finishing the proof. $t^k = r$ gives $\sqrt{\frac{\log_2 r}{\log_2 \log_2 r}} \log_2 t = \log_2 r$ so that $\log_2 t = \sqrt{\log_2 r \cdot \log_2 \log_2 r}$, so that $k = o(\log_2 t)$. $\square$

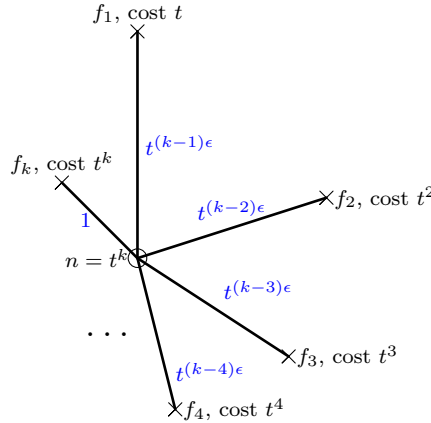


Figure 3: The star graph instance in Theorem 3. There are $r = t^k$ clients at the central vertex and no clients at any of the leaf vertices. Leaf vertex f_i has facility with cost t^i and is at a distance $t^{(k-i)\epsilon}$ from the central vertex. The distances are marked in blue.

3 Refinements

In this section, we will present our algorithms for weak and strong refinements to the facility location problem (see Section 1.1 for the definition of refinements). The key idea is to use the local relationship between approximate solutions, i.e., when two norms are relatively close, a good approximation for one is a good approximation for the other (see Lemma 2, Theorem 2). We will build on this idea to provide solution sets where the set of open facilities changes in a structured way, as required for strong and weak refinements.

We give 16-approximate weak refinements for arbitrary metrics in Theorem 4, Section 3.1.1. Strong refinements require a structural relationship between solutions for different norms not only in

terms of facilities, but also in terms of assignments. We give approximation algorithms for strong refinements for $P = [1, \infty]$ for arbitrary metrics in Section 3.1.2. We improve the approximation ratio for the special cases of line and tree metrics in Sections 3.2, 3.3 respectively.

From here on, we are primarily concerned by the order of our approximation ratios in terms of r . For convenience, we abuse the notation slightly and assume $\log_2 r \simeq \lceil \log_2 r \rceil$.

3.1 Arbitrary metric

We give approximation algorithms for weak and strong refinements for arbitrary metrics in this section. The main results in this section are Theorems 4, 5, and Corollary 2. Theorem 4 gives $O(1)$ -approximate weak refinement for all norms, i.e., the norm set $P = [1, \infty]$. Theorem 5 gives an $O(e^{3\sqrt{l}})$ -approximate strong refinement for any finite norm set $P = \{p_1, \dots, p_l\}$ and Corollary 2 gives a $\text{poly}(r^{\frac{1}{\sqrt{\log_2 r}}})$ -approximate strong refinement for the norm set $P = [1, \infty]$. Note that $e^{3\sqrt{l}}$ is *subexponential* in l and $r^{\frac{1}{\sqrt{\log_2 r}}}$ is *sublinear* in r : in fact, $r^{\frac{1}{\sqrt{\log_2 r}}} = 2^{\sqrt{\log_2 r}} = O(r^\delta)$ for all $\delta > 0$. These peculiar functions arise as bounds on the terms of a recurrence relation that occurs naturally in our problem (see Lemmas 4, 5).

3.1.1 Weak refinement

The key idea to get weak refinements is very simple: given norms $p < q$, we can augment an approximate solution $\mathcal{S}_p = (F_p, \theta_p)$ for the p -norm problem by including facilities F_q , where $\mathcal{S}_q = (F_q, \theta_q)$ is an approximate solution for the q -norm problem. Since $OPT_p \geq OPT_q$, the cost of the augmented solution $\mathcal{R}_p = (F_p \cup F_q, \theta_p)$ is at most twice the cost of $\mathcal{S}_p$, and the solution set $\{\mathcal{R}_p, \mathcal{S}_q\}$ forms a weak refinement for $P = \{p, q\}$.

This idea can be extended to larger norm sets. Theorem 2 allows us to pick a norm set of cardinality $\log_2 r$ to get a facility refinement for $P = [1, \infty]$.

Theorem 4. *Given any instance of the facility location problem, there exists a 16-approximate weak refinement for the norm set $[1, \infty]$. These weak refinements can be obtained in polynomial-time.*

Proof. Recall the norm set $\{p_0, \dots, p_t\}$ constructed in Theorem 2: $p_0 = 1$, $t = \lceil \log_2 \frac{OPT_1}{OPT_\infty} \rceil - 1$ and $OPT_{p_k} = \frac{OPT_{p_1}}{2^k}$ for all $k \in [t]$. Additionally, $p_{t+1} = \infty$.

We showed that a 4-approximate solution for the p_k -norm fair problem is an 8-approximate solution for the p -norm fair problem for all $p \in [p_k, p_{k+1}]$. Therefore, an α -approximate facility refinement for $P = \{p_0, \dots, p_t\}$ is a 2α -approximate facility refinement for $[1, \infty]$. We will obtain an 8-approximate facility refinement for P , implying the theorem.

For all $k \in [0, t]$, let $\mathcal{S}_k = (F_k, \theta_k)$ be a 4-approximate solution to the p_k -norm fair problem, obtained using Corollary 1. Define G_k as the union of all facilities from F_k through F_t , i.e., $G_k = \bigcup_{s \in [k, t]} F_s$. Then $G_0 \supseteq G_1 \supseteq \dots \supseteq G_t$. Further, define solution $\mathcal{R}_k = (G_k, \theta_k)$. Then,

$$\begin{aligned} C_{\mathcal{R}_k} &= C_{G_k} + C_{\theta_k} \leq \left(\sum_{s \in [k, t]} C_{F_s} \right) + C_{\theta_k} = \left(\sum_{s \in [k+1, t]} C_{F_s} \right) + (C_{F_k} + C_{\theta_k}) \\ &\leq \left(\sum_{s \in [k+1, t]} 4 OPT_{p_s} \right) + 4 OPT_{p_k} = \left(\sum_{s \in [k+1, t]} 4 \frac{OPT_{p_k}}{2^{s-k}} \right) + 4 OPT_{p_k} \end{aligned}$$

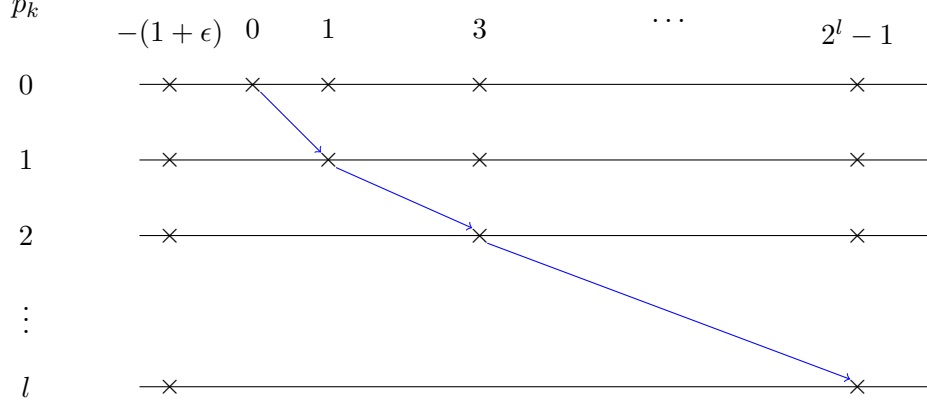


Figure 4: A toy example to illustrate that the greedy algorithm for strong refinements can reassign clients to far away facilities. The underlying metric is a line metric, and elements of sets $G_0, \dots, G_l$ are denoted as a cross at their respective locations, with a copy of the line for each $k \in [l]$ for clarity. For $\epsilon > 0$, the closest facility to $x = 0$ in G_1 is at $x = 1$, and so the greedy algorithm assigns $\Pi_1(0) = 1$. Further, the closest facility to $x = 1$ in G_2 is at $x = 3$, so that $\Pi_2(0) = 3$. Continuing this way, $\Pi_l(0) = 2^l - 1$. However, the closest facility to $x = 0$ in G_l is at $-(1 + \epsilon)$. For $\epsilon \simeq 0$, this reassignment gives a factor $\beta \simeq 2^l - 1$ for the client at $x = 0$.

$$= 4 \text{OPT}_{p_k} \left(\sum_{s \in [k, l]} \frac{1}{2^{s-k}} \right) \leq 4 \text{OPT}_{p_k} \left(\sum_{s \geq k} \frac{1}{2^{s-k}} \right) = 8 \text{OPT}_{p_k}.$$

Therefore, $\mathcal{R}_k$ is an 8-approximate solution for the p_k -norm fair problem, implying that the set $\{\mathcal{R}_0, \dots, \mathcal{R}_t\}$ is an 8-approximate facility refinement for P , and consequently a 16-approximate facility refinement for $[1, \infty]$. $\square$

3.1.2 Strong refinement

We proceed to discuss strong refinement. Given a finite $P = \{p_1, \dots, p_l\}$, we give a polynomial-time algorithm that gives an $O(e^{3\sqrt{l}})$ -approximate strong refinement for P . The first step is to get a weak refinement, as in Theorem 4. A weak refinement may not be a strong refinement, and so a natural question is if we can reassign the clients to facilities in each solution (i.e., for each $k \in [l]$) so that we get a strong refinement. Such a reassignment may increase the assignment cost for each $k \in [l]$. Notice that we do not open any facilities in any such reassignment. The key idea in this section is to give a reassignment such that the distance of each client from its assigned facility increases by at most a factor β , and this enables us to convert the bound of β into an approximate factor for strong refinement. We will show that $\beta = O(e^{3\sqrt{l}})$ for our algorithm. The main results for arbitrary metrics are given in Theorem 5 and Corollary 2, which we will improve upon this for the line metric in Section 3.2. Before we present the main algorithm in this section, we develop intuition using the greedy algorithm, and show why it fails.

Greedy algorithm and why it fails. From Theorem 4, we can obtain an $O(1)$ -weak refinement $\{(G_k, \Pi_k) : k \in [l]\}$ with $G_1 \supseteq \dots \supseteq G_l$. We denote by G_0 the set of clients and the set of facilities open for the 1-norm solution, i.e., $G_0 = D \cup G_1$ for convenience, so that $G_0 \supseteq G_1 \supseteq \dots \supseteq G_l$. Let us first consider a natural greedy algorithm: for each $j \in G_0$, assign $\Pi_1(j) = \arg \min_{f \in G_1} d(j, f)$, that is, the facility in G_1 closest to j . Call this facility f_1 . Under Π_2 , assign j to the facility in G_2 closest to f_1 in G_2 , i.e., $\Pi_2(j) = \arg \min_{f \in G_2} d(f_1, f)$, and so on. Recursively, for client j , assign

Algorithm 1 DISCOUNTEDLOOKAHEAD

- 1: Obtain solutions $\{\mathcal{S}_k = (F_k, \Pi_k) : k \in [l]\}$ such that $\mathcal{S}_k$ is a 4-approximation to the p_k -norm fair facility location problem.
- 2: Denote $G_k = \cup_{t \in [k, l]} F_t$ for $k \in [l]$
- 3: For any $k, t \in [0, l]$ and any $f \in G_k$, denote $h_t(f) = \arg \min_{h \in G_t} d(f, h)$
- 4: **for** $k = l - 1$ **to** 0 **do**
- 5: **for** each $f \in G_k$ **do**

$$s^* = \arg \min_{t \in [k+1, l]} \frac{d(f, h_t(f))}{\gamma^t},$$

where s^* is the least such number in case of ties. Let h be the corresponding facility, i.e., $h = h_{s^*}(f)$.

- 6: Set

$$\Pi_t^{(k)}(f) = \begin{cases} h & \text{if } t \in [k+1, s^*]; \\ \Pi_t^{(s^*)}(h) & \text{if } t \in [s^*+1, l] \end{cases}$$

- 7: **end for**

- 8: **end for**

- 9: Output solutions $(G_1, \Pi_1^{(0)}), \dots, (G_l, \Pi_l^{(0)})$
-

$\Pi_{k+1}(j) = \arg \min_{f \in G_{k+1}} d(\Pi_k(j), f)$ for all $k \geq 0$. It is not difficult to see that this algorithm produces a strong refinement, albeit with a higher approximation factor. Note that for higher k , this can lead to some clients being reassigned to facilities very far away; see Figure 4 for an example where the increase in the approximation factor can be very large for a client. Therefore, the greedy algorithm fails to provide a factor better⁴ than 2^l .

Our approach through DISCOUNTEDLOOKAHEAD. One can think of the greedy algorithm as looking ‘one step’ ahead: for each $k \in [0, l-1]$ and each $f \in G_k$, map $\sigma(f) = \arg \min_{h \in G_{k+1}} d(f, h)$. Then, for each client $j \in G_0$, the greedy algorithm assigns $\Pi_k(j) = \sigma^k(j)$. A natural question we consider here is if we can improve this algorithm by looking *more than* one step ahead at a time? Specifically, for client $j \in G_0$, instead of looking only at facility $f_1 = \arg \min_{f \in G_1} d(j, f)$, we can also look at facilities $h_k = \arg \min_{h \in G_k} d(j, h)$ for $k \in [l]$ (that is, h_k is the facility closest to j in G_k , so that $f_1 = h_1$). We pick some ‘discounting factor’ $\gamma > 1$ to be chosen later, and instead of assigning $\Pi_1(j) = h_1$, we assign $\Pi_{s^*}(j) = h_{s^*}$ where

$$s_* = \arg \min_{k \in [l]} \frac{d(j, h_k)}{\gamma^k}.$$

Since $G_k \supseteq G_{s^*}$ for all $k \leq s^*$, we can assign $\Pi_k(j) = h_{s^*}$, and for $k > s^*$, we can do this process recursively for $G_{s^*}, G_{s^*+1}, \dots, G_l$. Algorithm 1 does precisely this, and we show that this leads to a *subexponential* approximation factor in l .

For each $k \in [0, l-1]$ and each $t \in [k+1, l]$, the algorithm constructs an assignment $\Pi_t^{(k)} : G_k \rightarrow G_t$ from facilities at a lower level k to facilities at a higher level t . The final assignments on clients $G_0 = D$ are $\Pi_k^{(0)} : G_0 \rightarrow G_k$ for $k \in [l]$.

Lemma 3. *The assignments $\Pi_1^{(0)}, \dots, \Pi_l^{(0)}$ output by Algorithm 1 form a strong refinement.*

⁴We can prove that no client can be assigned *too* far away under the greedy algorithm. We can prove the bound $\beta \leq 2^l$, so that the greedy algorithm gives an $O(2^l)$ strong refinement.

Proof. First, we use induction on $l - k$ to prove the following: if $f \in G_t$, then $\Pi_s^{(k)}(f) = f$ for all $k < s \leq t$. When $k = l - 1$, this is trivially true since $d(f, f) = 0$. When $k < l - 1$, $s^* = s + 1$ and $h_{k+1}(f) = f$ in step 5 since $d(f, f) = 0$ and $f \in G_{k+1}$. By induction, $\Pi_t^{(k+1)}(f) = f$, and since by step 6, we have $\Pi_t^{(k)}(f) = \Pi_t^{(k+1)}(h_{k+1}(f))$, we get $\Pi_t^{(k)}(f) = f$.

For convenience, define $\Pi_k^{(k)} : G_k \rightarrow G_k$, where $\Pi_k^{(k)}(f) = f$ for all $f \in G_k$. We prove the following claim: for all $0 \leq k < s \leq t \leq l$ and for all (open) facilities $f \in G_k$ for $k \geq 1$ (and clients in G_0 for $k = 0$),

$$\Pi_t^{(k)}(f) = \Pi_t^{(s)} \left(\Pi_s^{(k)}(f) \right).$$

We prove this using induction on $l - k$. For $k = l - 1$, $s = t = l$, and by definition $\Pi_l^{(l)} \left(\Pi_l^{(k)}(f) \right) = \Pi_l^{(k)}(f)$.

Suppose $l - k > 1$, i.e., $k < l - 1$. As in Algorithm 1, let $s^* = \arg \min_{s' \in [k+1, l]} \gamma^{-s'} d(f, h_{s'}(f))$ and denote $h = h_{s^*}(f)$ for brevity in the proof that follows.

Case I: $s \leq t \leq s^*$. In this case, by step 6, $\Pi_t^{(k)}(f) = \Pi_s^{(k)}(f) = h$. Since $h \in G_t$, by our earlier claim, $\Pi_t^{(s)}(h) = h$, proving the desired equality.

Case II: $s < s^* \leq t$. Since $h \in G_{s^*}$, by our earlier claim we have $\Pi_{s^*}^{(s)}(h) = h$. Since $l - s^* < l - k$, using the induction hypothesis

$$\Pi_t^{(s)}(h) = \Pi_t^{(s^*)} \left(\Pi_{s^*}^{(s)}(h) \right) = \Pi_t^{(s^*)}(h).$$

By step 6, $\Pi_s^{(k)}(f) = h$ while $\Pi_t^{(k)}(f) = \Pi_t^{(s^*)}(h)$, so that

$$\Pi_t^{(s)} \left(\Pi_s^{(k)}(f) \right) = \Pi_t^{(s)}(h) = \Pi_t^{(s^*)}(h) = \Pi_t^{(k)}(f).$$

Case III: $s^* \leq s \leq t$. Since $l - s < l - k$, by the induction hypothesis,

$$\Pi_t^{(s^*)}(h) = \Pi_t^{(s)} \left(\Pi_{s^*}^{(s^*)}(f) \right). \quad (2)$$

In this case, $\Pi_t^{(k)}(f) = \Pi_t^{(s^*)}(h)$ and $\Pi_s^{(k)}(f) = \Pi_{s^*}^{(s^*)}(h)$, thus proving the desired equality and consequently the claim.

Facility sets $G_1 \supseteq \dots \supseteq G_l$ already form a weak refinement. To prove that assignments $\Pi_1^{(0)}, \dots, \Pi_l^{(0)}$, we need to show that for all $k \in [1, l - 1]$ and for all $f \in G_k$, there exists some $f' \in G_{k+1}$ such that each client assigned to f under $\Pi_k^{(0)}$ is assigned to f' under $\Pi_{k+1}^{(0)}$, that is:

$$\left\{ j \in G_0 : \Pi_k^{(0)}(j) = f \right\} \subseteq \left\{ j \in G_0 : \Pi_{k+1}^{(0)}(j) = f' \right\}.$$

We claim that $f' = \Pi_{k+1}^{(k)}(f)$ works.

Suppose $j \in G_0$ is such that $\Pi_k^{(0)}(j) = f$. Then j must have been assigned to f under $\Pi_k^{(0)}$ in one of the two cases in step 6. Let s^* be the corresponding index chosen when j is assigned.

If $k < s^*$, then $f = \Pi_k(j) = \Pi_{k+1}(j) = h_{s^*}(j)$. Since $h_{s^*}(j) \in G_{k+1}$, $\Pi_{k+1}^{(k)}(h_{s^*}(j)) = h_{s^*}(j)$.

If $k \geq s^*$, then $f = \Pi_k^{(0)}(j) = \Pi_k^{(s^*)}(h_{s^*}(j))$ and $\Pi_{k+1}^{(0)}(j) = \Pi_{k+1}^{(s^*)}(h_{s^*}(j))$. By eqn. (2),

$$\Pi_{k+1}^{(0)}(j) = \Pi_{k+1}^{(s^*)}(h_{s^*}(j)) = \Pi_{k+1}^{(k)} \left(\Pi_{s^*}^{(s^*)}(h_{s^*}(j)) \right) = \Pi_{k+1}^{(k)}(f).$$

□

Before we prove our main theorem for this section, we need the following technical lemmas; we include their proofs in Appendix C.

Lemma 4. *Given $\gamma > 1$ and some $l \in \mathbb{Z}_+$, consider the set of recursively defined numbers $\{u_{k,t} : t \in [l], k \in [0, t-1]\}$: for each $t \in [l]$, and for $k = t-1, \dots, 0$ define*

$$u_{k,t} = \begin{cases} \gamma^{l-t} & \text{if } k = t-1, \\ \max \{ \gamma^{l-t}, \max_{k < s < t} \gamma^{s-t} + u_{s,t}(1 + \gamma^{s-t}) \} & \text{if } k \in [0, t-2]. \end{cases} \quad (3)$$

Then $\max_{k,t} u_{k,t} \leq e^{\frac{2\gamma}{\gamma-1}} \gamma^{l-1}$.

Lemma 5. $\min_{\gamma > 1} e^{\frac{2\gamma}{\gamma-1}} \gamma^{l-1} \leq e^2 \cdot e^{3\sqrt{l}}$.

We are ready to prove the approximation guarantee for Algorithm 1:

Theorem 5. *Given a finite $P = \{p_1, \dots, p_l\}$ of norms, there is a polynomial-time algorithm that gives an $O(e^{3\sqrt{l}})$ -approximate strong refinement for P .*

Proof. Lemma 3 shows that Algorithm 1 gives a strong refinement for P . It remains to determine the choice of the parameter γ and prove the desired bound on the approximation ratio.

Assume without loss of generality that $p_1 < \dots < p_l$; we know that $OPT_{p_1} \geq \dots \geq OPT_{p_l}$. In step 2 of the algorithm,

$$C_{G_k} \leq \sum_{t \in [k,l]} C_{F_t} \leq 4 \sum_{t \in [k,l]} OPT_{p_t} \leq 4 \sum_{t \in [k,l]} OPT_{p_k} \leq 4l OPT_{p_k}.$$

If $\theta_k : D \rightarrow G_k$ assigns every client to its closest vertex in G_k , we have $C_{\theta_k} \leq C_{\Pi_k}$ since $G_k \supseteq F_k$. We will show that assignments $\Pi_k^{(0)}$ output by the algorithm satisfy $C_{\Pi_k^{(0)}} \leq O(e^{3\sqrt{l}}) C_{\theta_k}$, and therefore that

$$C_{(G_k, \Pi_k^{(0)})} = C_{G_k} + C_{\Pi_k^{(0)}} \leq 4l OPT_{p_k} + O(e^{3\sqrt{l}}) C_{\Pi_k} \leq (l + O(e^{3\sqrt{l}})) OPT_{p_k} = O(e^{3\sqrt{l}}) OPT_{p_k}.$$

To this end, it is sufficient to show that the assignments $\Pi_1^{(0)}, \dots, \Pi_l^{(l)}$ satisfy

$$d(j, \Pi_k^{(0)}(j)) \leq O(e^{3\sqrt{l}}) d(j, h_k(j)), \quad \forall j \in D. \quad (4)$$

Instead of proving (4), we prove the following stronger inductive statement:

$$d(f, \Pi_t^{(k)}(f)) \leq u_{k,t} d(f, h_t(f)), \quad \forall f \in G_k, \forall k \in [0, l-1], t \in [k+1, l]. \quad (5)$$

where u is the sequence defined in Lemma 4. Lemmas 4, 5 then imply (4).

We induct on $l - k$. Base case: $l - k = 1$. Consider any $f \in G_k$, and let s^* be the index in $[k+1, l]$ chosen in step 5. As in the algorithm, we denote $h = h_{s^*}(f)$. Then $s^* = l$, $h = h_l(f)$, and $\Pi_l^{(k)}(f) = h_l(f)$, so that (5) is trivially true.

Suppose $l - k > 1$. Consider any $f \in G_k$. We have two cases.

Case I: $t \leq s^*$. By definition, $\Pi_t^{(k)}(f) = h_{s^*}(f) := h$. By the choice of s^* , $\gamma^{-t} d(h_t(f), f) \geq \gamma^{-s^*} d(h, f)$, so that $d(h, f) \leq \gamma^{s^*-t} d(h_t(f), f) \leq \gamma^{l-t} d(h_t(f), f) \leq u_{k,t} d(h_t(f), f)$.

Case II: $t > s^*$. Since $s^* > k$, $l - s^* < l - k$, and so we have by the induction hypothesis that

$$d\left(h, \Pi_t^{(s^*)}(h)\right) \leq u_{s^*,t} d(h, h_t(h)) \leq u_{s^*,t} d(h, h_t(f)),$$

where the last inequality follows since by definition $h_t(h)$ is the closest facility to h in G_t .

Therefore, we have

$$\begin{aligned} d\left(f, \Pi_t^{(k)}(f)\right) &\leq d(f, h) + d\left(h, \Pi_t^{(k)}(f)\right) \\ &= d(f, h) + d\left(h, \Pi_t^{(s^*)}(h)\right) \\ &\leq d(f, h) + u_{s^*,t} \cdot d(h, h_t(f)) \\ &\leq d(f, h) + u_{s^*,t} \cdot (d(f, h) + d(f, h_t(f))). \end{aligned}$$

The first and final inequalities are triangle inequalities. The equality is by definition of $\Pi_t^{(k)}(f)$ and s^* .

Now, since $s^* = \arg \min_{t \in [k+1, l]} d(f, h_t(f))$, we have $d(f, h) \leq \gamma^{s^*-t} d(f, h_t(f))$, so that the above inequality becomes

$$d\left(f, \Pi_t^{(k)}(f)\right) \leq d(f, h_t(f)) \times \left(\gamma^{s^*-t} + u_{s^*,t} \cdot (\gamma^{s^*-t} + 1)\right).$$

Since

$$u_{k,t} \geq \max_{k < s < t} \left(\gamma^{s-t} + u_{s,t} \cdot (\gamma^{s-t} + 1)\right) \geq \gamma^{s^*-t} + u_{s^*,t} \cdot (\gamma^{s^*-t} + 1),$$

we get that

$$d\left(f, \Pi_t^{(k)}(f)\right) \leq d(f, h_t(f)) \times u_{k,t}.$$

□

Using Theorem 2, we can choose a finite norm set of size $l = \log_2 r$ to represent the norm set $P = [1, \infty]$ to get the following corollary:

Corollary 2. *There is a $\text{poly}(r^{\frac{1}{\sqrt{\log_2 r}}})$ -approximate strong refinement for $P = [1, \infty]$ for arbitrary metrics.*

3.2 Line metric

Consider now the special case when the underlying metric $\mathcal{M} = (X, d)$ is a line metric, i.e., $X = \mathbb{R}$. We denote by x_i the location of facility $i \in F$. In this section, we give an algorithm that produces an $O(\log_2 r)$ -approximate strong refinement for $P = [1, \infty]$. We prove the following general theorem for a finite norm set P , and obtain the result for $P = [1, \infty]$ as a corollary using Theorem 2:

Theorem 6. *Given a finite set of norms P , there is a polynomial-time algorithm that gives an $O(|P|)$ -approximate strong refinement for P for the line metric.*

Corollary 3. *There is a polynomial-time algorithm that gives an $O(\log_2 r)$ -approximate strong refinement for $P = [1, \infty]$ for the line metric.*

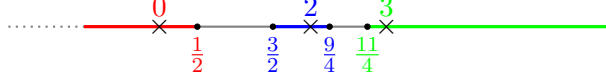


Figure 5: An example illustrating intervals I for $\alpha = 1/4$. There are three facilities, located at 0, 2, 3 respectively.

We proceed to develop the algorithm for $P = \{p_1, \dots, p_l\}$. We assign intervals to facilities instead of clients, with the understanding that each client in the interval is assigned to the corresponding facility. Assume without loss of generality that $p_1 < \dots < p_l$, so that $OPT_{p_1} \geq \dots \geq OPT_{p_l}$. Obtain a 4-approximate solution $\mathcal{S}_k = (F_k, \theta_k)$ for each $k \in [l]$. A weak refinement can be obtained in a manner that we have deployed before: by combining facilities at all higher norms for each p_k . That is, define $G_k = \cup_{t \in [k, l]} F_t$ and form new solutions

$$\mathcal{R}_k = (G_k, \theta_k).$$

We can bound the facility costs of $\mathcal{R}_k$ for each k :

$$C_{\mathcal{R}_k} = C_{G_k} + C_{\theta_k} \leq \left(\sum_{t \geq k} C_{F_t} \right) + C_{\theta_k} \leq 4l OPT_{p_k} + C_{\theta_k}. \quad (6)$$

We can reassign each θ_k so that each client j is mapped to its closest facility in G_k ; this does not increase the cost of the solutions $\mathcal{R}_k$.

We will give assignments $\Pi_k : D \rightarrow G_k$ so that $T = \{(G_k, \Pi_k) : k \in [l]\}$ forms a strong refinement and the assignment cost $C_{\Pi_k} \leq 2l C_{\theta_k}$. Combined with above and since $C_{\theta_k} \leq 4OPT_{p_k}$, this will imply that T forms a $12l$ -approximate strong refinement for P .

For notational convenience, we denote the facility $i \in G_k$ as $i^{(k)}$, and if it is also in $G_{k'}$ for some $k \neq k'$, we treat $i^{(k)}$ as different from $i^{(k')}$. Let $\alpha \in [0, 1/2]$ be a parameter that we fix later. For each facility, $i^{(k)}$, suppose $i_L^{(k)}$ is the facility to its immediate left in G_k and $i_R^{(k)}$ is the facility on its immediate right in G_k (if $i^{(k)}$ is the leftmost facility in G_k , we think of $i_L^{(k)}$ as lying at $-\infty$ and if it is the rightmost facility, we think of $i_R^{(k)}$ as lying at $+\infty$). Define

$$a_{i^{(k)}} = \alpha x_{i_L^{(k)}} + (1 - \alpha)x_{i^{(k)}}, \quad b_{i^{(k)}} = (1 - \alpha)x_{i^{(k)}} + \alpha x_{i_R^{(k)}}, \quad I_{i^{(k)}} = [a_{i^{(k)}}, b_{i^{(k)}}].$$

That is, $x_{i^{(k)}} - a_{i^{(k)}} = \alpha(x_{i^{(k)}} - x_{i_L^{(k)}})$ and $b_{i^{(k)}} - x_{i^{(k)}} = \alpha(x_{i_R^{(k)}} - x_{i^{(k)}})$.

It is useful to visualize this with an example: if $G_k = \{0, 2, 3\}$ with $x_0 = 0, x_2 = 2, x_3 = 3$, Figure 5 gives the intervals $I_{i^{(k)}}$ with $\alpha = \frac{1}{4}$. Note that $I_{0^{(k)}} = (-\infty, 1/2]$, $I_{2^{(k)}} = [3/2, 9/4]$, $I_{3^{(k)}} = [11/4, +\infty)$.

Here is an important idea for the algorithm: for any two consecutive facilities $i, h \in G_k$, if some Π_k assigns each client in $[x_i, b_i]$ to i and each client in $[a_h, x_h]$ to h , then irrespective of whether Π_k assigns clients in (b_i, a_h) to i or h , $C_{\Pi_k} \leq \frac{1}{\alpha} C_{\theta_k}$.

How do we ensure that such an assignment forms a strong refinement? For an example, see Figure 6 with just two norms $p_1 < p_2$. The facilities in G_1, G_2 are marked as crosses; denote the facility location at x as i_x . The intervals I for $\alpha = \frac{1}{4}$ are marked in blue and red respectively. As discussed, for any facility $i^{(k)}$, we want each client in $I_{i^{(k)}}$ to be assigned to $i^{(k)}$ under Π_k .

By definition of a strong refinement, all clients assigned to the facility $i_1^{(1)}$ at $x = 1$ in G_1 need to be assigned to a single facility in G_2 . Since $I_{i_1^{(1)}} = [3/4, 5/4]$ intersects $I_{i_0^{(2)}} = (-\infty, 1]$, the only option

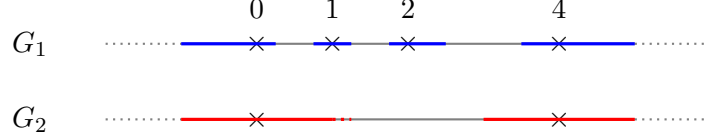


Figure 6: An example to illustrate our technique to produce a strong refinement on the line metric. Facilities in G_1 , G_2 are drawn as crosses and the corresponding intervals I are drawn in blue, red respectively. We require that all clients in interval $I_{i_1}^{(1)}$ for the facility location at $x = 1$ be assigned to a single facility in G_2 under Π_2 . Since the interval $I_{i_1}^{(1)} = [3/4, 5/4]$ intersects the interval $I_{i_0}^{(2)}$ for the facility i_0 at $x = 0$, we must expand the interval $I_{i_0}^{(2)}$ to include interval $I_{i_1}^{(1)}$ (in dotted red above).

is to assign all clients in $I_{i_1}^{(1)}$ to $i_0^{(2)}$ under Π_2 . Geometrically, this means expanding the interval $I_{i_0}^{(2)}$ to $I_{i_0}^{(2)} \cup I_{i_1}^{(1)} = (-\infty, 5/4]$. For facility $i_2^{(1)}$, since $I_{i_1}^{(1)}$ does not intersect either $I_{i_0}^{(2)}$ or $I_{i_4}^{(2)}$, all clients assigned to i_2 under Π_1 can be assigned to either $i_0^{(2)}$ or $i_4^{(2)}$ under Π_2 .

This can be done recursively up to level l . We need to make sure that this produces well-defined assignments $\Pi_1, \dots, \Pi_l$. A first step to ensure that requires that the intervals assigned to two different facilities in the same G_k do not intersect; i.e., no client is assigned to more than one facility in G_k . We show that for a small enough α , this can be achieved.

With some foresight, we choose $\alpha = \frac{1}{2l}$. We will make sure that assignments Π_k form a strong refinement for P . These assignments will be such that all of $I_{i^{(k)}}$ is assigned to $i^{(k)}$ for each facility $i^{(k)}$, so that $T = \{(G_k, \Pi_k) : k \in [l]\}$ will be a $12l$ -approximate strong refinement for P from eqn. (6).

We call our algorithm EXPANDINTERVALS, and it consists of three main steps. These have been separated into three subroutines for convenience:

1. Expand intervals I such that they form an appropriate hierarchical structure for partitions called *interval trees* which we shortly define (Algorithm 2, called GETHIERARCHYINTERVALTREE).
2. Rearrange the hierarchical structure so that it becomes easier to deal with unassigned intervals (Algorithm 3, called REARRANGEHIERARCHYINTERVALTREE).
3. Assign all unassigned intervals (Algorithm 4, called COMPLETEASSIGNMENT).

For convenience, we introduce a new level $l+1$ with $G_{l+1} = \{i_0\}$ where i_0 is an auxiliary node, and $I_{i_0}^{(l+1)} = \mathbb{R}$.

We want the intervals we assign to various facilities to form an interval tree structure, defined as follows:

Definition (Interval tree). A rooted tree H' is an interval tree with respect to intervals $A : V(H') \rightarrow \mathbb{R}$ if for vertices $u, v \in V(H')$:

1. if u is the parent of v in H' , then $A_v \subseteq A_u$, and
2. if u, v are siblings in H' then A_u, A_v are internally disjoint.

Algorithm 2 GETHIERARCHYINTERVALTREE

- 1: Set $\alpha = \frac{1}{2l}$
- 2: Set partial mapping $\sigma \leftarrow \emptyset$
- 3: **for** $k = 1$ to $l + 1$ **do**:
- 4: **for** each $i^{(k)} \in G_k$ **do**
- 5: Define $S_{i^{(k)}} = \{h^{(k')} : k' < k, h^{(k')} \text{ is unassigned under } \sigma, A_{h^{(k')}} \cap I_{i^{(k)}} \neq \emptyset\}$ and set

$$A_{i^{(k)}} \leftarrow I_{i^{(k)}} \cup \left(\bigcup_{h^{(k')} \in S_{i^{(k)}}} A_{h^{(k')}} \right)$$

- 6: Assign $\sigma(h^{(k')}) = i^{(k)}$ for all $h^{(k')} \in S_{i^{(k)}}$
 - 7: **end for**
 - 8: **end for**
 - 9: **return** σ and $A_{i^{(k)}}$ for all $k \in [l + 1]$, $i^{(k)} \in G_k$.
-

Algorithm 2 achieves precisely this aim. In the algorithm, we maintain a partial mapping $\sigma : \bigcup_{k \in [l]} G_k \rightarrow \bigcup_{k \in [2, l+1]} G_k$, initialized as $\sigma = \emptyset$. For any facility $i^{(k)} \in G_k$, mapping σ always maps $i^{(k)}$ to some $h^{(k')}$ with $k' > k$.

We show that Algorithm 2 forms an interval tree with respect to intervals A in Lemma 7. For all $k \in [l]$ and for all facilities $i^{(k)} \in G_k$, let $A_{i^{(k)}}$ output by Algorithm 2 be denoted $[c_{i^{(k)}}, d_{i^{(k)}}]$. We need the following structural lemma first:

Lemma 6. *For all $k \in [l]$, if $i^{(k)}$ and $h^{(k)}$ are consecutive facilities in G_k with $h^{(k)}$ to the right of $i^{(k)}$, then*

$$d_{i^{(k)}} - x_{i^{(k)}} \leq k\alpha x, \quad x_{h^{(k)}} - c_{h^{(k)}} \leq k\alpha x, \quad c_{h^{(k)}} - d_{i^{(k)}} \geq x(1 - 2k\alpha), \quad (7)$$

where $x = x_{h^{(k)}} - x_{i^{(k)}}$ is the distance between $i^{(k)}$ and $h^{(k)}$.

Proof. The third inequality is implied by the first two; we prove the first two inequalities by induction on k .

For $k = 1$, $A_{i^{(k)}} = I_{i^{(k)}}$ and $A_{h^{(k)}} = I_{h^{(k)}}$, so that $d_{i^{(k)}} = b_{i^{(k)}}$ and $c_{h^{(k)}} = a_{h^{(k)}}$, and the result follows by definition of intervals I .

Assume the result is true for all $k' < k$. Let

$$g^{(k_1)} = \arg \max_{h^{(k')} \in S_{i^{(k)}}} d_{h^{(k')}}.$$

Then, by step 5 of the algorithm, we have

$$d_{i^{(k)}} = \max \{b_{i^{(k)}}, d_{g^{(k_1)}}\}.$$

Since $k_1 < k$, we have that $i^{(k_1)}, h^{(k_1)} \in G_{k_1}$. By the induction hypothesis, then,

$$d_{g^{(k_1)}} - x_{g^{(k_1)}} \leq k_1\alpha(x_{g^{(k_1)}} - x_{i^{(k)}}), \quad x_{g^{(k_1)}} - c_{g^{(k_1)}} \leq k_1\alpha(x_{h^{(k)}} - x_{g^{(k_1)}}),$$

so that

$$d_{g^{(k_1)}} - c_{g^{(k_1)}} \leq k_1\alpha x.$$

Since $A_{i^{(k)}} \cap I_{h^{(k_1)}} \neq \emptyset$, $c_{g^{(k_1)}} \leq b_{i^{(k)}} = x_{i^{(k)}} + \alpha x$. Then we get that

$$d_{g^{(k_1)}} - x_{i^{(k)}} \leq (k_1 + 1)\alpha x \leq k\alpha x.$$

Since $d_{i^{(k)}} = \max\{b_{i^{(k)}}, d_{g^{(k_1)}}\}$, this proves the first inequality in (7). The second inequality is analogously proven. $\square$

Definition (Hierarchy graph). Define the directed graph $H = (V, E)$ with $V = \bigcup_{k \in [l+1]} G_k$ and $E = \{(i^{(k)}, h^{(k')}) : \sigma(h^{(k')}) = i^{(k)}\}$. This is simply a graphical representation of the function σ .

Observe that if $(i^{(k)}, h^{(k')}) \in E(H)$, then $k' < k$ by Algorithm 2.

Lemma 7. *The hierarchy graph H is an interval tree with respect to intervals A output by Algorithm 2.*

Proof. For each $k \in [1, l+1]$, we first prove the following (stronger) statement: let H_k denote the subgraph of H induced by the vertex set $\bigcup_{k' \in [k]} H_{k'}$. Then each component of H_k is an interval tree with respect to intervals A , and furthermore, for two components C_1, C_2 of H_k with roots r_1, r_2 , then A_{r_1} and A_{r_2} are internally disjoint.

We prove this by induction on k . For $k = 1$, each component of H_1 is simply an isolated vertex and further, $A_{i^{(1)}}$ and $A_{h^{(1)}}$ are simply disjoint intervals for $i \neq h$, implying the claim.

Assume now that $k > 1$ and that the claim is true for all $k' < k$; we prove it for k .

Let C be a component of H_k rooted at $i^{(k')}$. We first prove that C is an interval tree. If $k' < k$, then C is also a component of H_{k-1} and therefore C is an interval tree by the induction hypothesis. Suppose $k' = k$. The only parent-child pairs present in $H_k \cap C$ but not in $H_{k-1} \cap C$ are $i^{(k)}$ and one of its children. And the only sibling pairs present in $H_k \cap C$ but not in $H_{k-1} \cap C$ are two children of $i^{(k)}$.

Let $g_1^{(k_1)}$ be some child of $i^{(k)}$ in C . By definition of H , $\sigma(g_1^{(k_1)}) = i^{(k)}$, and therefore $A_{g_1^{(k_1)}} \subseteq A_{i^{(k)}}$ by step 5 of the algorithm. Let $g_2^{(k_2)}$ be some child of $i^{(k)}$ in C . Since $i^{(k)} \notin V(H_{k-1})$, both $g_1^{(k_1)}, g_2^{(k_2)}$ are the roots of their corresponding components in H_{k-1} . By the induction hypothesis, $A_{g_1^{(k_1)}} \cap A_{g_2^{(k_2)}} = \emptyset$ are internally disjoint. This proves that C is an interval tree.

Let C_1, C_2 be two components in H_k , rooted at $i_1^{(k_1)}, i_2^{(k_2)}$ respectively. we claim that $A_{i_1^{(k_1)}}$ and $A_{i_2^{(k_2)}}$ are internally disjoint.

If both $k_1, k_2 < k$, then $A_{i_1^{(k_1)}}$ and $A_{i_2^{(k_2)}}$ are internally disjoint by the induction hypothesis.

If $k_1 = k_2 = k$, then Lemma 6 then implies that $A_{i_1^{(k_1)}}$ and $A_{i_2^{(k_2)}}$ are internally disjoint.

Suppose $k_1 = k$ and $k_2 < k$. Since $i_2^{(k_2)}$ is a root of a component in H_k , $i_2^{(k_2)}$ must have been unassigned at the end of iteration k in loop 3. Therefore, since $\sigma(i_2^{(k_2)}) \neq i_1^{(k_1)}$, $i_2^{(k_2)} \notin S_{i_1^{(k_1)}}$ and therefore $A_{i_1^{(k_1)}} \cap A_{i_2^{(k_2)}} = \emptyset$ by step 5.

Finally, since $A_{i_0^{(l+1)}} = I_{i_0^{(l+1)}} = \mathbb{R}$, by step 5, $H = H_{l+1}$ is a tree with root i_0 , and therefore an interval tree by our claim. $\square$

Notice also that by step 5 of Algorithm 2, $A_{i^{(k)}} \supseteq I_{i^{(k)}}$ for each $i^{(k)}$. We aim to eventually assign all of $A_{i^{(k)}}$ to $i^{(k)}$ under Π_k , and therefore this observation will be crucial in bounding the increase in assignment costs, as previously discussed. We record this as a lemma:

Algorithm 3 REARRANGEHIERARCHYINTERVALTREE

```

1: for  $k = l + 1$  to 2 do
2:   for each facility  $i^{(k)} \in G_k$  do
3:     Let the children of  $i^{(k)}$  in  $H$  be  $C = \{i_1^{(k_1)}, \dots, i_t^{(k_t)}\}$ 
4:     for each  $i_s^{(k_s)} \in G_1 \cup \dots \cup G_{k-2}$  do
5:       Let  $i_p^{(k-1)}$  be the facility in  $C \cap G_{k-1}$  closest to  $i_s^{(k_s)}$ 
6:       Remove edge  $(i^{(k)}, i_s^{(k_s)})$  from  $H$  and add edge  $(i_p^{(k-1)}, i_s^{(k_s)})$  to  $H$ 
7:       Update  $A_{i_p^{(k-1)}} \leftarrow \text{conv}(A_{i_p^{(k-1)}} \cup A_{i_s^{(k_s)}})$   $\triangleright \text{conv}(A)$  denotes the convex hull of set  $A$ 
8:     end for
9:   end for
10: end for
11: return  $H$  and intervals  $A_{i^{(k)}}$  for all  $k \in [l + 1]$ ,  $i^{(k)} \in G_k$ 

```

Lemma 8. For each $k \in [l]$ and each $i^{(k)} \in G_k$, the interval $A_{i^{(k)}}$ output by Algorithm 2 satisfies $A_{i^{(k)}} \supseteq I_{i^{(k)}}$.

At this point, we are close to getting a strong refinement: we can assign $A_{i^{(k)}}$ to $i^{(k)}$, and the interval tree structure (Lemma 7) implies that this will give a strong refinement. Lemma 8 bounds the assignment costs for these assignments. The only trouble is that the sets $\{A_{i^{(k)}} : i^{(k)} \in G_k\}$ may not cover the whole of $\mathbb{R}$, i.e., it is possible that $\bigcup_{i^{(k)} \in G_k} A_{i^{(k)}} \subsetneq \mathbb{R}$, so that the assignment is *partial*. We aim to remedy this situation going ahead.

For each facility $i^{(k)}$, let $T(i^{(k)})$ denote the sub-tree in H rooted at $i^{(k)}$. Note that the children of $i^{(k)}$ in H are facilities in $G_1 \cup \dots \cup G_{k-1}$. To produce a strong refinement, we need that the children of $i^{(k)}$ be facilities in G_{k-1} only. For this, we modify H by expanding some intervals $A_{i^{(k)}}$ and rearranging some edges in H in Algorithm 3.

For $1 < k \leq l$, $i^{(k)}$ always has at least one child (namely $i^{(k-1)}$) in G_{k-1} . For $i_0^{(l+1)}$, each facility in G_l is a child.

Lemma 9. The tree H is an interval tree with respect to intervals A output by Algorithm 3. Further, each child of facility $i^{(k)}$ in H is a facility in G_{k-1} for all $k \in [l + 1]$, $i^{(k)} \in G_k$.

Proof. We claim that H stays an interval tree at all times in Algorithm 3.

$A_{i^{(k)}}$ is updated in the algorithm only if $i^{(k)}$ has a parent $h^{(k+1)}$ in H initially, and not updated thereafter. From step 7 then, each $A_{i^{(k)}}$ is still an interval.

Suppose $g_1^{(k_1)}, \dots, g_t^{(k_t)}$ are the children of $h^{(k+1)}$ reassigned to $i^{(k)}$. Then $A_{i^{(k)}} \supseteq A_{g_s^{(k_s)}}$ for each $s \in [t]$ since H is an interval tree at all previous steps of the algorithm, and therefore $A_{i^{(k)}}$ stays a subset of $A_{h^{(k+1)}}$ after it is updated. Further, the update to $A_{i^{(k)}}$ in step 7 ensures that it is a superset of $A_{g_s^{(k_s)}}$ for each $s \in [t]$.

Further, if some child $g^{(k')}$ of $h^{(k+1)}$ is reassigned to $i^{(k)}$ in the algorithm, then all children of $h^{(k+1)}$ between $g^{(k')}$ and $i^{(k)}$ are also assigned to $i^{(k)}$ by the condition in step 5. This implies that if $i'^{(k)}$ is another child of $h^{(k)}$ in G_k , then, $A_{i^{(k)}}$ does not intersect $A_{i'^{(k)}}$ after taking the convex hull in step 7. $\square$

Next, we finally assign all intervals that are still unassigned in Algorithm 4, while maintaining the interval tree structure for H . Once this is done, the assignments $\Pi_1, \dots, \Pi_l$ output by the

Algorithm 4 COMPLETEASSIGNMENT

```
1: Set  $u = -\infty, v = +\infty$ 
2: Run REFINEBOUNDARY( $H, u, v$ )
3: for each  $i^{(k)} \in G_1 \cup \dots \cup G_l$  do
4:   Denote  $B_{i^{(k)}} = [u_{i^{(k)}}, v_{i^{(k)}}]$ , assign  $\Pi_k(B_{i^{(k)}}) = i^{(k)}$ 
5: end for
6: return assignments  $\Pi_1, \dots, \Pi_l$ 
```

Algorithm 5 REFINEBOUNDARY($T(i^{(k)}), u_{i^{(k)}}, v_{i^{(k)}}$)

```
1: input: a subtree  $T(i^{(k)})$  of  $H$ , left boundary point  $u_{i^{(k)}}$  and right boundary point  $v_{i^{(k)}}$ 
2: if  $k = 1$  then
3:   return
4: else
5:   Let  $i_1^{(k-1)}, \dots, i_t^{(k-1)}$  be the children of  $i^{(k)}$  in  $T(i^{(k)})$  ordered from left to right
6:   Set  $u_{i_1^{(k-1)}} = u_{i^{(k)}}$  and  $v_{i_t^{(k-1)}} = v_{i^{(k)}}$ 
7:   for  $s = 1$  to  $t - 1$  do
8:     Select an arbitrary point  $w_s$  satisfying
           
$$d_{i_s^{(k-1)}} \leq w_s \leq c_{i_{s+1}^{(k-1)}}$$

9:     Set  $v_{i_s^{(k-1)}} = w_s, u_{i_{s+1}^{(k-1)}} = w_s$ 
10:  end for
11:  for  $s = 1$  to  $t$  do
12:    Run RefineBoundary( $T(i_s^{(k-1)}), u_{i_s^{(k-1)}}, v_{i_s^{(k-1)}}$ )
13:  end for
14: end if
```

algorithm are well-defined functions, so that each client is assigned to exactly one facility in G_k under Π_k . Additionally, with the interval tree structure on H , assignments $\Pi_1, \dots, \Pi_l$ form a strong refinement.

Note that the sets $\{B_{i^{(k)}} : i^{(k)} \in G_k\}$ partition $\mathbb{R}$ for each $k \in [l]$; this is easy to show by induction. This shows that assignments $\Pi_1, \dots, \Pi_l$ are well defined.

Similarly, simple induction proves that H is an interval tree for intervals B , so that the assignments $\Pi_1, \dots, \Pi_l$ form a strong refinement.

From Lemma 8, $A_{i^{(k)}} \supseteq I_{i^{(k)}}$ for each facility $i^{(k)}$. Note also that each $B_{i^{(k)}} \supseteq A_{i^{(k)}}$ by the choice of w_s in Algorithm 5. Therefore, we have that $B_{i^{(k)}} \supseteq I_{i^{(k)}}$ for each facility $i^{(k)}$.

Recall that to prove the approximation guarantee, it is enough to prove that for each k , the assignment costs satisfy

$$C_{\Pi_k} \leq 2l C_{\theta_k}$$

Indeed, we will show that the cost of each client j goes up at most by a factor of $2l = \frac{1}{\alpha}$ when reassigned from θ_k to Π_k .

Note that

- For each $k \in [l]$ and for each client j , $\Pi_k(j)$ is either the facility in G_k on the immediate left

of j or on the immediate right of j .

- For each $i^{(k)}$, each client in $I_{i^{(k)}}$ is assigned to $i^{(k)}$ under Π_k .

For two consecutive facilities $i^{(k)}, h^{(k)} \in G_k$, clients j between $x_{i^{(k)}}$ and $x_{h^{(k)}}$ such that $\frac{x_j - x_{i^{(k)}}}{x_j - x_{h^{(k)}}} \leq \alpha$ are precisely the clients in $I_{i^{(k)}}$. These are all assigned to $i^{(k)}$ under Π_k and therefore for these clients $C_{\Pi_k} = C_{\theta_k}$. Similarly, the same relation holds for clients $j \in I_{h^{(k)}}$ between $x_{i^{(k)}}$ and $x_{h^{(k)}}$.

For all other clients j between $x_{i^{(k)}}$ and $x_{h^{(k)}}$, assigning j to either $i^{(k)}$ or $h^{(k)}$ increases the cost by at most $1/\alpha$, that is, $C_{\Pi_k} \leq \frac{1}{\alpha} C_{\theta_k}$. This finishes the proof.

3.3 Tree metric

Consider the case when the underlying metric is induced by a tree, i.e., there is a tree $M = (T, E, w)$ with vertices T , edges E , and nonnegative edge weights $w : E \rightarrow \mathbb{R}$ and the distance $d(j, j')$ is the sum of edge weights on the unique path from j to j' . Under the assumptions that the facility costs are uniform and that a facility can be opened anywhere on the tree, we give an $O(|P|)$ -approximate strong refinement for a finite norm set P (Theorem 7) and an $O(\log r)$ -approximate strong refinement for all norms, i.e., norm set $P = [1, \infty]$ (Corollary 4).

Given a tree T' , a vertex $v \in V(T')$ is called a *branch vertex* if $\deg_{T'}(v) \geq 3$. Here is the key idea for extending the algorithm for line metric to the tree metric: for a finite norm set $P = \{p_1, \dots, p_l\}$ with $p_1 < \dots < p_l$, suppose we get a weak refinement for P using Theorem 4, with facility sets $G_1 \supseteq \dots \supseteq G_l$. Suppose we open a new facility at each branch vertex of the tree induced by G_k for each $k \in [l]$. We can show that these new facility sets also form a weak refinement. Interestingly, we can now treat the path joining facility $i^{(k)}, h^{(k)} \in G_k$ as a line! This allows us to call the strong refinement algorithm for the line metric, EXPANDINTERVALS as a subroutine.

We have a simple lemma to bound the number of branch vertices first; this lemma allows us to open facilities at branch vertices under the assumption of uniform facility costs:

Lemma 10. *For any tree, the number of branch vertices is strictly less than the number of leaves in the tree.*

Proof. Given a tree T' , let s_1 be the number of leaf vertices in T' , s_2 be the number of degree 2 vertices in T' , and s be the number of branch vertices in T' . By the handshaking lemma, we have $2(s_1 + s_2 + s - 1) = \sum_{v \in V(T')} \deg_{T'}(v) \geq s_1 + 2s_2 + 3s$, implying $s_b \leq s_1 - 2$. $\square$

Given a tree T' and a set of vertices $S \subseteq V(T')$, let T'_S denote the subtree of T' induced by all vertices in S and those in the paths connecting any two vertices in S . We will use algorithm EXPANDINTERVALS from Section 3.2 as a subroutine. Recall that EXPANDINTERVALS takes as input facility sets $\mathbb{R} \supseteq G_1 \supseteq \dots \supseteq G_t$ for some $t \geq 1$ (where each G_k , $k \in [t]$ is nonempty and finite) and produces assignments $\Pi_k : \mathbb{R} \rightarrow G_k$ for each $k \in [t]$ such that ⁵

- for each $k \in [t]$, the sets $\{\Pi_k^{-1}(i) : i \in G_k\}$ are internally disjoint intervals that partition $\mathbb{R}$,
- For each $k \in [t]$, and consecutive facilities $i, h \in G_k$, with h to the right of i , the interval $[i, i + \alpha(h - i)] \in \Pi_k^{-1}(i)$ and $[h - \alpha(h - i), h] \in \Pi_k^{-1}(h)$ for $\alpha = \frac{k}{2t}$, and

⁵We use the standard set theoretic notation $\Pi_k^{-1}(i) = \{x \in \mathbb{R} : \Pi_k(x) = i\}$.

Algorithm 6 BRANCHANDLINEARIZE($T, G_1, \dots, G_l$)

```
1: Initialize empty mappings  $\Pi_1, \dots, \Pi_l = \emptyset$ 
2: For convenience, denote  $G_{l+1} = \emptyset$ 
3: For all  $k \in [l+1]$ , add all branch vertices in  $T_{G_k}$  to  $G_k$ ; call this set  $G'_k$ 
4: for  $k = l$  to 1 do
5:   for each path  $p$  between two consecutive vertices of  $G'_k$  in  $T_{G_k} \setminus T_{G_{k+1}}$  do
6:     Solve strong partition problem for  $p$ , i.e., treat path  $p$  as a line with lengths determined
       by edge weights  $w$ , and obtain assignments
       
$$\Pi'_1, \dots, \Pi'_k \leftarrow \text{EXPANDINTERVALS}(G'_1, \dots, G'_k),$$

       where each assignment  $\Pi'_s : V(p) \rightarrow V(p) \cap G'_s, s \in [k]$ 
7:     for  $s \in [k]$  do
8:       Update assignments  $\Pi_s \leftarrow \Pi_s \cup \Pi'_s$ 
9:     end for
10:  end for
11:  for each component  $C$  of  $T \setminus T_{G_k}$  do
12:    Let  $v$  be the vertex that connects  $C$  to  $T_{G_k}$ 
13:    Assign  $\Pi_k(j) = \Pi_k(v)$  for all  $j \in V(C)$ 
14:  end for
15: end for
16: Return solutions  $\{\mathcal{T}_k = (G'_k, \Pi_k) : k \in [l]\}$ 
```

- For each $k \in [t-1]$ and each $i \in G_k$, there exists some $i' \in G_{k+1}$ such that $\Pi_k^{-1}(i) \subseteq \Pi_{k+1}^{-1}(i')$.

Definition (Definition). Given a tree T' and a vertex set $S \subseteq V(T')$, two vertices u, v are called *consecutive vertices of S* if the unique path in T' joining u, v does not contain any other vertex of S .

Notice that the edges in tree T'_S can be uniquely partitioned into paths between pairs of consecutive vertices of S . We use this fact to decompose trees into paths between facilities.

Theorem 7. *Given a set of norms $P = \{p_1, \dots, p_l\}$ and an instance of the fair facility location problem with a tree metric, Algorithm 6 gives an $O(l)$ -approximate strong refinement for P in polynomial time, assuming uniform facility costs and that a facility can be opened at any vertex of the tree.*

Proof. For each $k \in [l]$, let $\mathcal{S}_k = (F_k, \theta_k)$ be 4-approximate solution to the p_k -norm fair facility location on T . As usual, for all $k \in [l]$, one can obtain solutions $\mathcal{R}_k = (G_k, \theta_k)$ where $G_k = \cup_{t \in [k, l]} F_t$ so that

$$C_{\mathcal{R}_k} = C_{G_k} + C_{\theta_k} \leq 4l C_{\mathcal{S}_k} + C_{\theta'_k} \leq 4l OPT_k + C_{\theta_k}.$$

Algorithm 6 takes as input this $O(l)$ -approximate weak refinement for T , with the facility sets $G_1 \supseteq G_2 \supseteq \dots \supseteq G_l$.

We first show that the algorithm produces well-defined assignments $\Pi_1, \dots, \Pi_l$, i.e., that each Π_k is a function from $V(T)$ to G'_k . We use induction on $l - k$. When $k = l$, $T_{G_l} \setminus T_{G_{l+1}} = T_{G_l}$ since $T_{G_{l+1}} = \emptyset$ by definition (see step 2). Since the edges in T_{G_l} can be partitioned into paths between pairs of consecutive vertices of G'_l , step 8 determines the assignment $\Pi_l(j)$ for all $j \in V(T_{G_l})$. For vertices $j \in V(T) \setminus V(G_l)$, $\Pi_l(j)$ is determined by step 13.

Suppose $k < l$. Notice that for any $t > k$, a branch vertex of G_t is also a branch vertex of G_k , so that $G'_k \subseteq G'_t$. Therefore, edges in $T_{G_k} \setminus T_{G_{k+1}}$ can be partitioned into disjoint paths between some pairs of vertices in G_k . For vertices $j \in V(T_{G_{k+1}})$, $\Pi_k(j)$ is determined before loop 4 reaches k . For vertices $j \in V(T_{G_k}) \setminus V(G_{k+1})$, $\Pi_k(j)$ is determined in step 8. For vertices $j \in V(T) \setminus V(G_k)$, $\Pi_k(j)$ is determined by step 13.

Next, we prove that solutions $\{\mathcal{T}_k = (G'_k, \Pi_k)\}$ form a strong refinement. Since $G'_k \supseteq G'_{k+1}$ for all $k \in [l-1]$, these solutions form a weak refinement. Pick any $i \in G'_k$, we claim that there exists some $i' \in G'_{k+1}$ such that $\Pi_k^{-1}(i) \subseteq \Pi_{k+1}^{-1}(i')$.

Case I: $i \notin G'_{k+1}$. Let $C_1, \dots, C_t$ be the components of $T \setminus i$. First, notice that at most 2 of $C_1, \dots, C_t$ can contain a vertex of G_{k+1} (otherwise i would be a branch vertex of G_{k+1} and would be present in G'_{k+1}). Therefore, by definition of G'_{k+1} , at most 2 of $C_1, \dots, C_t$ can contain a vertex of G'_{k+1} . Further, since $i \notin G'_{k+1}$ and $G'_{k+1} \neq \emptyset$, at least one of $C_1, \dots, C_t$ contains a vertex of G'_{k+1} .

Case I.A: Exactly one of $C_1, \dots, C_t$ contains a vertex of G'_{k+1} , say C_1 . Then $T_{G'_{k+1}} \subseteq C_1$. Let h be the nearest vertex of i in $G'_k \cap V(C_1)$, and let the path from i to h be called p . Then $T_{G'_{k+1}} \subseteq C'$, where C' is the component of $C_1 \setminus p$ containing h . Notice that $\{i\} \cup (C_1 \setminus C') \cup C_2 \cup \dots \cup C_t$ is a subtree of some component of $T \setminus T_{G_{k+1}}$. Then, by step 13, each vertex j in this subtree satisfies $\Pi_{k+1}(j) = \Pi_{k+1}(h)$. Further, since $\Pi_k^{-1}(i)$ is a subtree of T and $h \notin \Pi_k^{-1}(i)$, $\Pi_k^{-1}(i)$ is a subtree of $\{i\} \cup (C_1 \setminus C') \cup C_2 \cup \dots \cup C_t$. Therefore, the choice $i' = \Pi_{k+1}(h)$ works.

Case I.B: Exactly two of $C_1, \dots, C_t$ contain a vertex of G'_{k+1} , say C_1, C_2 . Then, there exist $h_1 \in V(C_1) \cap G'_{k+1}$, and $h_2 \in V(C_2) \cap G'_{k+1}$ such that i is on the path p from h_1 to h_2 and there is no other vertex of G'_{k+1} on p . Step 8 gets a strong refinement on path p . Assume without loss of generality that $\Pi_{k+1}(i) = h_1$. Since we have a strong refinement on p , we get $V(p) \cap \Pi_k^{-1}(i) \subseteq V(p) \cap \Pi_{k+1}^{-1}(h_1)$. Further, for each $j \in V(C_3), \dots, V(C_t)$, $\Pi_{k+1}(j) = \Pi_{k+1}(i) = h_1$ by step 13. Since $\Pi_k^{-1}(i) \subseteq V(p) \cup V(C_3) \cup V(C_3) \cup \dots \cup V(C_t)$, this implies that $\Pi_k^{-1}(i) \subseteq \Pi_{k+1}^{-1}(h_1)$.

Case II: $i \in G'_{k+1}$. We claim that for all components C of $T \setminus i$, $\Pi_k^{-1}(i) \cap V(C) \subseteq \Pi_{k+1}^{-1}(i) \cap V(C)$.

If $V(C) \cap G'_{k+1} = \emptyset$, then by step 13, $\Pi_{k+1}(j) = i$ for all $j \in V(C)$, and therefore, $\Pi_k^{-1}(i) \cap V(C) \subseteq \Pi_{k+1}^{-1}(i) \cap V(C)$.

If C contains some vertex from G'_{k+1} , let h be the closest such vertex to i . By definition of G'_{k+1} , the path p from i to h has no other vertex in G'_{k+1} . Since the strong partition problem is solved on p in step 8 in loop $k+1$, $\Pi_k^{-1}(i) \cap V(p) \subseteq \Pi_{k+1}^{-1}(i) \cap V(p)$. Furthermore, let i_1 be the closest vertex from G'_k to i on p (there is such a vertex since $h \in V(p)$). Let p' be the subpath of p from i to i_1 . Let C' be a component of $C - p'$ attached to some vertex $f \in V(p')$. If $\Pi_k(f) = i$, then $\Pi_{k+1}(f) = i$ since $\Pi_k^{-1}(i) \cap V(p) \subseteq \Pi_{k+1}^{-1}(i) \cap V(p)$. Therefore, if $\Pi_k(j) = i$ for all $j \in C'$, then $\Pi_{k+1}(j) = i + 1$. This implies the claim that $\Pi_k^{-1}(i) \cap V(C) \subseteq \Pi_{k+1}^{-1}(i) \cap V(C)$.

Finally, we prove that each $\mathcal{T}_k$ is an $O(l)$ -approximation for the p_k -norm fair facility location problem on T . Notice that step 3 introduces new facilities to solutions $\mathcal{R}_k$ and potentially increases the facility costs. However, since we assume uniform facility costs and since the set of leaves of T_{G_k} is a subset of G_k , Lemma 10 implies that the facility costs are not increased by a factor of more than 2, or that

$$C_{G'_k} \leq 2C_{G_k} \leq 8l \text{OPT}_k.$$

Further, we will show that assignment costs do not increase by a factor more than $2l$, so that overall

$$C_{\mathcal{T}_k} = C_{G'_k} + C_{\Pi_k} \leq 8l \text{OPT}_k + 2l C_{\theta_k} \leq 8l \text{OPT}_k + 8l \text{OPT}_k = 16l \text{OPT}_k.$$

To bound the increase in assignment costs, it is enough to show the following: for each $j \in V(T)$ and each $k \in [l]$,

$$\frac{d(j, \Pi_k(j))}{\min_{i \in G'_k} d(j, i)} \leq 2l.$$

If j lies on some path p connecting two vertices $i_1, i_2 \in G'_k$, then by step 8 and the strong partition for line metric EXPANDINTERVALS on p , the above inequality holds.

If j is not on such a path, let C be the component of $T \setminus T_{G_k}$ containing j , and let C be connected to T_{G_k} at vertex f . If $f \in G'_k$, then $f = \arg \min_{i \in G'_k} d(j, i)$ and we are done. Otherwise, $\deg_{T_{G_k}}(f) = 2$, and so let i_1, i_2 be the nearest branch vertices of T_{G_k} to f . Without loss of generality, assume i_1 is closer to f than i_2 ; then i_1 is closer to j than i_2 . We have from step 13 that $\Pi_k(j) = \Pi_k(f)$ and from step 8 that $\Pi_k(f)$ is either i_1 or i_2 . By the strong refinement EXPANDINTERVALS on path from i_1 to i_2 , we also have that

$$\frac{d(f, \Pi_k(f))}{d(f, i_1)} \leq 2l.$$

Therefore,

$$\frac{d(j, \Pi_k(j))}{d(j, i_1)} = \frac{d(j, \Pi_k(f))}{d(j, i_1)} = \frac{d(j, f) + d(j, \Pi_k(f))}{d(j, f) + d(f, i_1)} \leq \max\left(\frac{d(j, f)}{d(j, f)}, \frac{d(f, \Pi_k(f))}{d(f, i_1)}\right) \leq 2l.$$

□

Finally, we can use Theorem 2 to obtain the following corollary for norm set $P = [1, \infty]$:

Corollary 4. *Given an instance of the fair facility location problem with a tree metric, there is an $O(\log r)$ -approximate strong refinement for $P = [1, \infty]$ in polynomial time, assuming uniform facility costs and that a facility can be opened at any vertex of the tree.*

4 Trade-offs between opening and connection costs

In the section, we discuss trade-offs between opening and connection costs as a function of p by considering a weighting coefficient for the connection cost. We characterize the monotonicity of OPT as a function of p in terms of this weighting coefficient (8). We also provide an alternate model for fair facility location motivated by this analysis. We provide analogous results for the number of solutions and refinements for this proposed new model (Corollary 5, Theorems 9, 10). We also briefly comment on the related k -clustering problem (Section 4.2).

Consider the following simple example: the underlying metric is the line segment $[0, 1]$. There are n clients at $x = 0$; each client is the only one in its group. There are two facilities, at $x = 0$ and $x = 1$. The facility at $x = 0$ has opening cost $\sqrt{n}$ and the facility at $x = 1$ has opening cost 1.

The optimal 1-norm solution opens the facility at $x = 0$ with total cost $\sqrt{n}$. Note that the assignment cost is 0.

This is, however, a bad approximation for ∞ -norm objective. Indeed, it is easy to verify that the *only* $O(1)$ -approximate solution for ∞ -norm is to open the facility at $x = 1$, with the total cost 2. However, each client is now at a greater distance from its nearest open facility. Thus, fairness is being achieved by making everyone worse off.

There is an explanation for this behavior: for a given solution, as norm p increases, the assignment cost decreases while the facility cost stays the same. Therefore, there is higher incentive to minimize the facility cost as opposed to the assignment cost for higher p .

4.1 Normalized fair facility location

We can introduce a normalizing factor to balance the facility and assignment costs as p varies: for a solution $\mathcal{S} = (F', \Pi)$ and client groups $D_1, \dots, D_r$, we have the assignment cost vector

$$V_\Pi = \left(\frac{1}{|D_1|} \sum_{j \in D_1} d(j, \Pi(j)), \dots, \frac{1}{|D_r|} \sum_{j \in D_r} d(j, \Pi(j)) \right) \in \mathbb{R}^r.$$

With a nonnegative *normalizing factor* $g(p; r)$, we define the p -norm cost function to be

$$C_{\mathcal{S}} = C_{F'} + g(p; r) \|V_\Pi\|_p,$$

and the objective is to minimize this cost function across solutions.

A following question is the choice of the function $g(p; r)$. We first characterize the monotonicity of OPT as a function of p , which then leads to a natural choice of $g(p; r)$ that resolves the aforementioned unwanted behavior.

Theorem 8. 1. *OPT is a nonincreasing function of p if $g(p; r)$ is a nonincreasing function of p . Further, if $g(p; r)$ is not a nonincreasing function of p , then there exist problem instances where OPT is not a nonincreasing function of p .*

2. *OPT is a nondecreasing function of p if $r^{\frac{1}{p}} g(p; r)$ is a nondecreasing function of p . Further, if $r^{\frac{1}{p}} g(p; r)$ is not a nondecreasing function of p , then there exist problem instances where OPT is not a nondecreasing function of p .*

Proof. 1. Suppose $g(p; r)$ is a nonincreasing function of p . We show that $OPT_q \leq OPT_p$ for all norms $q > p$ for all problem instances with r groups. Given such a problem instance, let F_p, V_p respectively denote the set of open facilities and the assignment cost vector for the optimal solution for the p -norm problem. Then,

$$\begin{aligned} OPT_q &\leq F_p + g(q; r) \|V_p\|_q \\ &\leq F_p + g(p; r) \|V_p\|_q \\ &\leq F_p + g(p; r) \|V_p\|_p \\ &= OPT_p. \end{aligned}$$

The first inequality holds by definition of OPT_q , the second inequality holds since g is a nonincreasing function, and the final inequality holds by Lemma 1 and since g is nonnegative.

Suppose $g(p; r)$ is not nonincreasing, then there exist norms $p, q \in [1, \infty]$ with $p < q$ and $g(p; r) < g(q; r)$. Consider the line segment $[0, 1]$. There is a single facility at $x = 0$ with opening cost 0, with $r - 1$ clients at $x = 0$ and 1 client at $x = 1$. The only possible solution is to open the facility at $x = 0$, with $OPT_p = 0 + g(p; r) \times 1 = g(p; r)$ and similarly $OPT_q = g(q; r)$. Since $g(p; r) < g(q; r)$, we have $OPT_p < OPT_q$, so that OPT is not nonincreasing in p .

2. Suppose $r^{\frac{1}{p}} g(p; r)$ is a nondecreasing function of p . We show that $OPT_q \geq OPT_p$ for all norms $q > p$ for all problem instances with r groups. Given such a problem instance, let F_q, V_q respectively denote the set of open facilities and the assignment cost vector for the optimal solution for the q -norm problem. Then,

$$OPT_p \leq F_q + g(p; r) \|V_q\|_p$$

$$\begin{aligned}
&\leq F_q + g(p; r) r^{\frac{1}{p} - \frac{1}{q}} \|V_q\|_q \\
&\leq F_p + g(q; r) \|V_q\|_q \\
&= OPT_q.
\end{aligned}$$

The first inequality holds by definition of OPT_p , the second inequality holds by Lemma 1 and since g is nonnegative, and the final inequality holds since $r^{\frac{1}{p}}g(p; r)$ is a nondecreasing function.

Suppose $r^{\frac{1}{p}}g(p; r)$ is not nondecreasing, then there exist norms $p, q \in [1, \infty]$ with $p < q$ and $r^{\frac{1}{p}}g(p; r) > r^{\frac{1}{q}}g(q; r)$. Consider the star graph with r leaf vertices and unit distances between the central vertex and any any leaf. There is a single facility at $x = 0$ with opening cost 0. There is one client each at each of the leaf vertices, with each client in a singleton group. The only possible solution is to open the facility at $x = 0$, with $OPT_p = 0 + g(p; r) \times \|(1, \dots, 1)\|_p = r^{\frac{1}{p}}g(p; r)$ and similarly $OPT_q = r^{\frac{1}{q}}g(q; r)$. Since $r^{\frac{1}{p}}g(p; r) > r^{\frac{1}{q}}g(q; r)$, we have $OPT_p > OPT_q$, so that OPT is not nondecreasing in p . $\square$

This suggests $g(p; r) = r^{1 - \frac{1}{p}}$ as a natural choice for the normalizing factor. Lemma 1, which gives a fundamental relationship between different p -norms in Cartesian spaces suggests that the factor $r^{\frac{1}{p} - \frac{1}{q}}$ can offset the decrease in assignment cost as the norm is increased. We can normalize this with respect to the 1-norm and recover our choice of the normalizing factor.

Formally, we define the *normalized p -norm fair facility location problem* as an instance of the facility location problem with objective

$$\min_{(F', \Pi)} \sum_{i \in F'} c_i + r^{1 - \frac{1}{p}} \left\| \left(\frac{1}{|D_s|} \sum_{j \in D_s} d(j, \Pi(j)) \right)_{s \in [r]} \right\|_p,$$

that is, the problem seeks to minimize across all solutions the sum of facility costs and the p -norm of the vector of group distances weighted by $r^{1 - 1/p}$.

With our notation for facility and assignment costs, the cost for solution $\mathcal{S} = (F', \Pi)$ is $C_{\mathcal{S}} = C_{F'} + r^{1 - \frac{1}{p}} C_{\Pi}$. Notice that for $p = 1$ and singleton groups (i.e., each client is in a separate group), this reduces to the classic facility location problem.

Theorem 1 immediately implies that this problem is 4-approximable:

Corollary 5. *There is a polynomial-time algorithm that gives a 4-approximation for the normalized p -norm fair facility location problem for any $p \in [1, \infty]$.*

As with our original model, we can establish more structure across approximate solutions for different p -norms. Theorem 8 already tells us that OPT_p is a nondecreasing function of p . Analogous to Theorem 2, we can once again get a portfolio of $\lceil \log_2 r \rceil$ solutions such that for each norm $p \in [1, \infty]$, one of these solutions is an 8-approximation to the p -norm fair problem. We omit the proof as it is similar to the proof of Theorem 2.

Theorem 9. *Given any instance of the facility location problem, there exists an 8-approximate solution set of cardinality at most $\lceil \log_2 r \rceil$ for the normalized fair facility location problem. This solution set can be obtained in polynomial time.*

The advantage of the normalized problem is most significant for refinements. Since OPT_p is an increasing function of p , we expect that more facilities open as norm p increases, as opposed to

less facilities being open for our original model. In terms of fairness, this implies that fairness is achieved (i.e. the norm increases) by opening more facilities instead of closing them. This motivates the definition of *increasing refinements*:

Definition (Increasing refinements). We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty facility set $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}$. We are given a metric space $\mathcal{M} = (X, d)$, client set $D \subseteq X$, and non-empty facility set $F \subseteq X$ with non-negative facility costs $c : F \rightarrow \mathbb{R}$. Furthermore, suppose we are given non-empty client groups $D_1, \dots, D_r$ that partition D . Given a set of norms $P \subseteq [1, \infty]$, a solution set $S = \{\mathcal{S}_p = (F_p, \Pi_p) : p \in P\}$ is called

1. a *weak increasing refinement* if for all $p, q \in P$ with $p \leq q$, $F_p \subseteq F_q$; that is, the facilities form a subset structure.
2. a *strong increasing refinement* if for all $p, q \in P$ with $p \leq q$ and for each facility $i \in F_q$, there is some facility $i' \in F_p$ such that all clients assigned to i under Π_q are assigned to i' under Π_p , that is,

$$\{j \in D : \Pi_q(j) = i\} \subseteq \{j \in D : \Pi_p(j) = i'\}.$$

If for each norm $p \in P$, the solution $\mathcal{S}_p$ is an α -approximation for the p -norm fair problem, then we say that the refinement is an α -approximate increasing refinement.

Simple changes to our algorithms lead to analogous results for increasing refinements for the normalized fair facility location problem:

Theorem 10. 1. *There is 16-approximate increasing weak refinement for the norm set $[1, \infty]$ for the normalized fair facility location problem.*

2. *There is a $\text{poly}(r^{\frac{1}{\sqrt{\log_2 r}}})$ -approximate strong increasing refinement for the norm set $[1, \infty]$ for the normalized fair facility location problem for arbitrary metrics.*
3. *There is an $O(\log_2 r)$ -approximate strong increasing refinement for the norm set $[1, \infty]$ for the normalized fair facility location problem on the line metric.*
4. *There is an $O(\log_2 r)$ -approximate strong increasing refinement for the norm set $[1, \infty]$ for the normalized fair facility location problem on the tree metric, assuming uniform facility cost and that a facility can be opened anywhere on the tree.*

These increasing refinements can be obtained in polynomial time.

4.2 k -clustering

We can also consider the related clustering model where at most k facilities can be opened and the objective is to minimize the p -norm of the assignment vector. This generalizes k -median ($p = 1$), k -means ($p = 2$), and k -center ($p = \infty$) clustering problems. Several $O(1)$ -approximations are known for the p -norm k -clustering problem [25].

Analogous to Theorem 2, we can prove that OPT is a nonincreasing function of p and that at most $\lceil \log_2 r \rceil$ solutions are required such that for any p , one of these solutions is a good solution. If all of these solutions are combined, we immediately get that there is a solution with $O(k \log r)$ open facilities such that it is an $O(1)$ -approximation for all p -norms simultaneously; this is already known ([24, 25]).

5 Hierarchical facility location problem

In this section, we extend the algorithms for refinements to the hierarchical facility location problem defined in Section 1.1. We give approximation algorithms for various metrics; the results are presented in Theorem 11.

Given an instance of the hierarchical facility location problem with l levels, 4-approximations for the individual facility location problem at each level $k \in [l]$ can be obtained using Theorem 1 when the function f is sublinear and polynomial-time oracles for f and ∇f are available. Suppose $\mathcal{S}_k = (F_k, \theta_k)$ is a 4-approximation to the level k facility location problem (i.e., with facility costs $c_k \in \mathbb{R}_+$).

Note that since $c_1 \leq \dots \leq c_l$, we must have $OPT_1 \leq \dots \leq OPT_l$, where OPT_k is the value of the optimal solution to the level k problem. This allows us to get a subset structure in facility sets as in Theorem 4. That is, let $G_k = \cup_{t \in [k, l]} F_t$ and consider solutions $\mathcal{R}_k = (G_k, \theta_k)$ for the level k problem, $k \in [l]$. Solutions $\mathcal{R}_k$ satisfy constraint (1) in the definition of hierarchical facility location. We will show that $C_{\mathcal{R}_k} \leq 4l C_{\mathcal{S}_k} \leq 16l OPT_k$ for all k .

For any $t \geq k$, since $\mathcal{S}_t$ is a 4-approximation to the level t problem and since $\mathcal{S}_k$ is a feasible solution to the level t problem, we have

$$C_{\mathcal{S}_t} = c_t |F_t| + C_{\Pi_t} \leq 4 OPT_t \leq 4 (c_t |F_k| + C_{\Pi_k}),$$

implying that $|F_t| \leq 4|F_k| + \frac{4}{c_t} C_{\Pi_k}$. Therefore, using the monotonicity assumption $c_k \leq c_t$ for all $t \geq k$, we get

$$\begin{aligned} C_{\mathcal{R}_k} &= c_k |G_k| + C_{\Pi_k} \\ &\leq c_k \sum_{t \geq k} |F_t| + C_{\Pi_k} \\ &= C_{\mathcal{S}_k} + c_k \sum_{t > k} |F_t| \\ &\leq C_{\mathcal{S}_k} + c_k \sum_{t > k} \left(4|F_k| + \frac{4}{c_t} C_{\Pi_k} \right) \\ &\leq C_{\mathcal{S}_k} + \sum_{t > k} 4 (c_k |F_k| + C_{\Pi_k}) \\ &\leq 4l C_{\mathcal{S}_k}. \end{aligned}$$

If we can change the assignments $\Pi_k, k \in [r]$ so that they satisfy constraint (2) in the definition, we will get a feasible solution to hierarchical facility location problem. We claim that our algorithms for refinements are sufficient for this reassignment: indeed, none of the strong refinement algorithms DISCOUNTEDLOOKAHEAD, EXPANDINTERVALS, and BRANCHANDLINEARIZE use the norm structure of p -norms for reassigning clients. If the reassignment increases the cost of Π_k by a factor at most β for each k , we get that $\mathcal{R}_k$ is an $O(\beta l)$ -approximation for the level k problem, and therefore the set of solutions $R = \{\mathcal{R}_k : k \in [l]\}$ forms an $O(\beta l)$ -approximation for the hierarchical facility location problem. Therefore, we have the following theorem:

Theorem 11. *Given an instance of the hierarchical facility location problem with l levels and a differentiable sublinear function f with polynomial-time oracles for f and ∇f ,*

1. *there is a polynomial-time algorithm that is an $\tilde{O}(e^{3\sqrt{l}})$ -approximation for arbitrary metrics,*

2. there is a polynomial-time algorithm that is an $O(l^2)$ -approximation for the line metric, and
3. there is a polynomial-time algorithm that gives an $O(l^2)$ -approximation for the tree metric, assuming that a facility can be opened anywhere on the tree.

6 Open questions

A natural open question is about the existence of a solution set with cardinality lesser than $\lceil \log_2 r \rceil$ (Theorem 2, Section 2) that is constant-factor approximate for $P = [1, \infty]$. In particular, given any instance of the fair facility location problem, what is the correct order of the smallest number of solutions required: is it $\theta(\sqrt{\log_2 r})$, or $\theta(\log_2 r)$, or something between the two?

Another open question is whether the subexponential factor $e^{3\sqrt{l}}$ for strong refinement in Theorem 5 (Section 3.1) can be reduced to a polynomial factor. We believe that this should be true (potentially by searching a larger local neighborhood instead of a one-step for the Greedy algorithm):

Conjecture 1. *Given a finite set $P = \{p_1, \dots, p_l\}$ of norms, there is a polynomial-time algorithm that gives a $\text{poly}(l)$ -approximation to the strong refinement problem for arbitrary metrics.*

This can potentially (but not necessarily) lead to an improvement for the corresponding factor for the hierarchical facility location problem in Section 5. For strong refinements, it is natural to ask if the algorithm for line metric generalizes to Cartesian spaces $\mathbb{R}^d$, $d > 1$, after perhaps losing an additional factor that depends on d (2^d , for instance).

Finally, it is natural to ask the basic question about the approximability of the general facility location problem (see Theorem 1): can it be approximated to a factor better than 4 in polynomial time? Several improved approximation factors are known for the classic facility location problem [45], including the 1.488-approximation algorithm of Li. Most of these algorithms crucially use the linearity of the objective, and therefore do not generalize.

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A Proof of Theorem 1

We can formulate the problem as an integer program: for each facility $i \in F$, variable $y_i \in \{0, 1\}$ indicates whether facility i is opened and for each client $j \in D$, variable $x_{ij} \in \{0, 1\}$ indicates whether j has been assigned to i . The feasible region is

$$\sum_{i \in F} x_{ij} \geq 1, \quad \forall j \in D,$$

$$\begin{aligned} x_{ij} &\leq y_i, & \forall i \in F, j \in D, \\ x_{ij}, y_i &\in \{0, 1\}, & \forall i \in F, j \in D, \end{aligned}$$

and the objective is to minimize the function $\sum_{i \in F} c_i y_i + f\left(\left(\sum_{i \in F} x_{ij} d(i, j)\right)_{j \in D}\right)$.

In the relaxation, we allow variables x, y to take values in $[0, 1]$:

$$\min_{x, y} \sum_{i \in F} c_i y_i + f\left(\left(\sum_{i \in F} x_{ij} d(i, j)\right)_{j \in D}\right) \quad \text{subject to} \quad (\text{CP})$$

$$\sum_{i \in F} x_{ij} \geq 1, \quad \forall j \in D, \quad (8)$$

$$x_{ij} \leq y_i, \quad \forall i \in F, j \in D, \quad (9)$$

$$0 \leq x, y. \quad (10)$$

Let us denote the underlying polytope by P and the objective function by $\bar{f} : P \rightarrow \mathbb{R}$. Since P has a polynomial number of constraints, the Ellipsoid algorithm (see [7]) can be used to obtain the following theorem:

Theorem 12. *For a differentiable convex function $f : \mathbb{R}^D \rightarrow \mathbb{R}$, an optimal solution to (CP) can be obtained in a polynomial number of calls to f and ∇f .*

For the classical facility location problem with a linear objective, Shmoys *et al.* [41] use the filtering technique first introduced by Lin and Vitter [36] to round a fractional solution to (CP), giving a 4-approximation. We show that their rounding algorithm generalizes to a much larger class of convex functions:

Lemma 11. *If f is a sublinear function, then a fractional solution (x, y) to (CP) can be rounded to an integral solution (x^*, y^*) with cost $\bar{f}(x^*, y^*) \leq 4\bar{f}(x, y)$ in polynomial time.*

This, along with Theorem 12 implies Theorem 1 for the general facility location problem and the corollary for p -norm socially fair facility location problem.

We present the algorithm now and then proceed to prove Lemma 11.

Proof of Lemma 11. Consider the solution $(\bar{x}, \bar{y})$ output by FILTER. Since $\bar{x} \leq \frac{1}{\alpha} x$ and $\bar{y} = \frac{1}{\alpha} y$, it satisfies the feasibility constraints (9) and (10). Further,

$$\sum_{i \in F} \bar{x}_{ij} = \sum_{i: d_{ij} \leq d_j(\alpha)} \frac{1}{\alpha} x_{ij} = \frac{1}{\alpha} \sum_{i: d_{ij} \leq d_j(\alpha)} x_{ij} \geq \frac{1}{\alpha} \times \alpha = 1$$

by definition of $d_j(\alpha)$. Therefore, $\bar{x}, \bar{y}$ is feasible.

Furthermore

$$\sum_{i \in F} d_{ij} x_{ij} \geq \sum_{i: d_{ij} \geq d_j(\alpha)} d_{ij} x_{ij} \geq d_j(\alpha) \sum_{i: d_{ij} \geq d_j(\alpha)} x_{ij} \geq d_j(\alpha)(1 - \alpha). \quad (11)$$

Let the solution (x^*, y^*) output by Algorithm ROUND. For the facility i_k and client j_k chosen in iteration k of loop 5, by step 7 and since $\sum_{i \in F} \bar{x}_{ij_k} \geq 1$,

$$c_{i_k} \leq c_{i_k} \sum_{i \in F} \bar{x}_{ij_k} \leq \sum_{i \in N_G(j_k)} c_i \bar{x}_{ij_k} \leq \sum_{i \in N_G(j_k)} c_i \bar{y}_i.$$

Algorithm 7 FILTER(x, y)

```

1: Set  $\alpha = \frac{1}{4}$ 
2: for  $j \in D$  do
3:   Let  $S_j = \{i \in F : x_{ij} > 0\}$ 
4:   Let  $\sigma$  be the permutation arranging facilities in  $S_j$  such that  $d_{\sigma(1)j} \leq d_{\sigma(2)j} \leq \dots$ 
5:   Let  $i_j = \arg \min_k \sum_{k' \in [k]} x_{\sigma(k')j} \geq \alpha$ , and let  $d_j(\alpha) = d_{\sigma(i_j),j}$ 
6: end for
7: for  $i \in F$  do
8:   Set  $\bar{y}_i \leftarrow \frac{1}{\alpha} y_i$ 
9:   for  $j \in D$  do
10:    Set

$$\bar{x}_{ij} = \begin{cases} \frac{1}{\alpha} x_{ij} & \text{if } d_{ij} \leq d_j(\alpha), \\ 0 & \text{otherwise.} \end{cases}$$

11:   end for
12: end for
13: return  $(\bar{x}, \bar{y})$ 

```

Furthermore, since each client in set $N_G^2(j_k)$ is assigned to i_k and then removed from G , the client sets $N_G(j_1), N_G(j_2), \dots$ are disjoint. Along with the above, this implies

$$\sum_k c_{i_k} \leq \sum_k \sum_{i \in N_G(j_k)} c_i \bar{y}_i \leq \sum_{i \in F} c_i \bar{y}_i = \frac{1}{\alpha} \sum_{i \in F} c_i y_i. \quad (12)$$

Suppose client j was assigned to i_k . Then $j \in N_G^2(j_k)$ and by step 6, $d_j(\alpha) \geq d_{j_k}(\alpha)$. Since $j \in N_G^2(j_k)$, there exists some $i \in N_G(j) \cap N_G(j_k)$. By metric property of d ,

$$d_{i_k j} \leq d_{i_k j_k} + d_{i j_k} + d_{i j}.$$

Further, by step 10 and since $\bar{x}_{ij}, \bar{x}_{ij_k}, \bar{x}_{i_k j_k} > 0$, we must have $d_{ij} \leq d_j(\alpha)$, $d_{ij_k} \leq d_{j_k}(\alpha)$ and $d_{i_k j_k} \leq d_{j_k}(\alpha)$, so that

$$d_{i_k j} \leq 2d_{j_k}(\alpha) + d_j(\alpha) \leq 3d_j(\alpha).$$

Eqn. (11) then implies that

$$d_{i_k j} \leq 3d_j(\alpha) \leq \frac{3}{1-\alpha} \sum_{i \in F} x_{ij} d_{ij},$$

or that

$$\sum_{i \in F} x_{ij}^* d_{ij} \leq \frac{3}{1-\alpha} x_{ij} d_{ij}.$$

Since f is nondecreasing and sublinear,

$$f\left(\left(\sum_i x_{ij}^* d_{ij}\right)_{j \in D}\right) \leq f\left(\left(\frac{3}{1-\alpha} \sum_i x_{ij} d_{ij}\right)_{j \in D}\right) = \frac{3}{1-\alpha} f\left(\left(\sum_i x_{ij} d_{ij}\right)_{j \in D}\right).$$

By eqn. (12),

$$\sum_{i \in F} c_i y_i^* = \sum_k c_{i_k} \leq \frac{1}{\alpha} \sum_{i \in F} c_i y_i.$$

Together, for $\alpha = \frac{1}{4}$, this implies that $\bar{f}(x^*, y^*) \leq 4f(x, y)$. $\square$

Algorithm 8 ROUND($\bar{x}, \bar{y}$)

```
1: Initialize  $D' \leftarrow D$ 
2: Form a bipartite graph  $G$  with vertex sets  $D', F'$  with  $ji \in E(G)$  iff  $\bar{x}_{ij} > 0$  for all  $i \in F, j \in D$ 
3: For  $j \in D$ , denote by  $N_G(j) = \{i \in F : \bar{x}_{ij} > 0\}$  the neighborhood of  $j$  in  $G$  and by  $N_G^2(j)$  the set of all  $j' \in D$  that share a neighbor with  $j$  in  $G$ , i.e., the set of all  $j' \in D$  with  $N_G(j') \cap N_G(j) \neq \emptyset$ 
4:  $k = 0$  ▷ Loop counter
5: while  $D' \neq \emptyset$  do
6:   Choose  $j_k = \arg \min_{j \in D'} g_j$ 
7:   Choose  $i_k = \arg \min_{i \in N_G(j_k)} f_i$ 
8:   Open  $i_k$  and assign all  $j \in N_G^2(j_k)$  to  $i_k$ 
9:    $D' \leftarrow D' \setminus (N_G^2(j_k) \cup \{j_k\})$ , update  $G$ 
10: end while
11: Set  $y_i^* = 1$  for all opened facilities  $i$ , set  $x_{ij}^* = 1$  if  $j$  was assigned to  $i$ , and set all other variables to 0
12: return  $(x^*, y^*)$ 
```

B Missing proofs from Section 2

Proof of Lemma 1. The left hand side bound is standard; we omit the proof. ⁶

To see the other bound, note that $q/p \geq 1$ and $\frac{1}{q/p} + \frac{1}{q/(q-p)} = 1$. Holder's inequality then gives

$$\|x\|_p^p = \sum_{i \in [r]} |x_i|^p \times 1 \leq \left(\sum_{i \in [r]} |x_i|^{p \times \frac{q}{p}} \right)^{\frac{p}{q}} \cdot \|\mathbf{1}_r\|_{q/(q-p)} = \|x\|_q^p r^{\frac{q-p}{q}} = \|x\|_q^p \left(r^{\frac{1}{p} - \frac{1}{q}}\right)^p,$$

implying $\|x\|_p \leq r^{\frac{1}{p} - \frac{1}{q}} \|x\|_q$. □

Proof of Lemma 2. Let OPT_p, OPT_q denote the corresponding optimum values. Let $\mathcal{S}_p = (F_p, \Pi_p)$ be an α -approximate solution to the p -norm fair problem, and let C_p denote the assignment cost vector for it (so that the p -norm assignment cost of Π_p is $\|C_p\|_p$). Let $\mathcal{S}_q^* = (F_q^*, \Pi_q^*)$ be an optimum solution to the q -norm fair problem, with assignment cost vector C_q^* .

We have from definition that $C_{F_p} + \|C_p\|_p \leq \alpha OPT_p$. Now,

$$\begin{aligned} C_{F_p} + \|C_p\|_q &\leq C_{F_p} + \|C_p\|_p \\ &\leq \alpha OPT_p \\ &\leq \alpha (C_{F_q^*} + \|C_q^*\|_p) \\ &\leq \alpha (C_{F_q^*} + r^{\frac{1}{p} - \frac{1}{q}} \|C_q^*\|_q) \\ &\leq \alpha r^{\frac{1}{p} - \frac{1}{q}} (C_{F_q^*} + \|C_q^*\|_q) \\ &= \alpha r^{\frac{1}{p} - \frac{1}{q}} OPT_q. \end{aligned}$$

where the first inequality follows from Lemma 1 and since $q > p$, the second inequality is true since s_p is α -approximate, the third inequality follows from definition, the fourth inequality follows from

⁶One way to see this is to show that the derivative of $f(p) = \frac{1}{p} \sum_i |x_i|^p$ with respect to p is nonpositive for $p \geq 1$.

Lemma 1, the fifth inequality is true since $\frac{1}{p} - \frac{1}{q} > 0$, and the final equality is true since $\mathcal{S}_q^*$ is an optimal solution to the q -norm fair problem. Therefore, $\mathcal{S}_p$ is a $\alpha r^{\frac{1}{p} - \frac{1}{q}}$ -approximate solution to q -norm fair problem.

Similarly let $\mathcal{S}_q = (F_q, \Pi_q)$ be an α -approximate solution to the q -norm fair problem with cost vector C_q . Then, using the fact that $OPT_p \geq OPT_q$,

$$\begin{aligned} C_{F_q} + \|C_q\|_p &\leq C_{F_q} + r^{\frac{1}{p} - \frac{1}{q}} \|C_q\|_q \\ &\leq r^{\frac{1}{p} - \frac{1}{q}} (C_{F_q} + \|C_q\|_q) \\ &\leq \alpha r^{\frac{1}{p} - \frac{1}{q}} OPT_q \\ &\leq \alpha r^{\frac{1}{p} - \frac{1}{q}} OPT_p. \end{aligned}$$

Therefore, $\mathcal{S}_q$ is an $\alpha r^{\frac{1}{p} - \frac{1}{q}}$ -approximation to the p -norm fair problem □

C Missing proofs from Section 3

Proof of Lemma 4. First, note that for any $t \in [1, l]$, since $\gamma > 1$, we have

$$u_{t-1,t} \leq \dots \leq u_{1,t} \leq u_{0,t},$$

so that it is enough to prove that $\max_t u_{0,t} \leq e^{\frac{2\gamma}{\gamma-1}} \gamma^{l-1}$. Further, since $u_{t-1,t} = \gamma^{l-t}$, this implies

$$u_{k,t} = \max_{k \leq s \leq t} \gamma^{s-t} + u_{s,t}(1 + \gamma^{s-t}) \quad \forall k \in [0, t-2].$$

First, a change of indices is in order: if we define $w_{k,t} = u_{t-1-k,t}$ for $k \in [0, t-1]$, we get $w_{0,t} = \gamma^{l-t}$ and

$$w_{k,t} = \max_{s \in [0, k-1]} \gamma^{-(s+1)} + w_{s,t}(1 + \gamma^{-(s+1)}) \quad \forall k \in [1, t-1].$$

This makes it clear that in fact

$$w_{k,t} = \gamma^{-k} + w_{k-1,t}(1 + \gamma^{-k}) \quad \forall k \in [1, t-1].$$

Finally, we bound $w_{t-1,t} = u_{0,t}$. Since $w_{0,t} = \gamma^{l-t} \geq 1$, notice that

$$\frac{w_{k,t}}{w_{k-1,t}} \leq \frac{\gamma^{-k}}{w_{k-1,t}} + 1 + \gamma^{-k} \leq 1 + 2\gamma^{-k}.$$

Therefore, taking product from $k = 1$ to $t-1$, we get

$$\frac{w_{t-1,t}}{w_{0,t}} \leq \prod_{k=1}^{t-1} (1 + 2\gamma^{-k}).$$

Denote $P_t = \prod_{k=1}^{t-1} (1 + 2\gamma^{-k})$. Using the standard bound $1 + x \leq \exp(x)$,

$$\prod_{k=1}^{t-1} (1 + 2\gamma^{-k}) \leq \prod_{k=1}^{t-1} \exp(2\gamma^{-k}) = \exp\left(2 \sum_{k=1}^{t-1} \gamma^{-k}\right) \leq \exp\left(2 \sum_{k=1}^{\infty} \gamma^{-k}\right) = \exp\left(\frac{2\gamma}{\gamma-1}\right).$$

Therefore,

$$u_{0,t} = w_{t-1,t} \leq \exp\left(\frac{2\gamma}{\gamma-1}\right) w_{0,t} = \exp\left(\frac{2\gamma}{\gamma-1}\right) \gamma^{l-1}.$$

□

Proof of Lemma 5. Choose $\gamma = 1 + \frac{1}{\sqrt{l}}$. Then $e^{\frac{2\gamma}{\gamma-1}} = e^{2\sqrt{l}(1+\frac{1}{\sqrt{l}})} = e^{2\sqrt{l}} \cdot e^2$. Also,

$$\gamma^{l-1} < \gamma^l = \left(1 + \frac{1}{\sqrt{l}}\right)^l \leq \left(e^{\frac{1}{\sqrt{l}}}\right)^l = e^{\sqrt{l}},$$

implying that $e^{\frac{2\gamma}{\gamma-1}} \gamma^{l-1} \leq e^2 \cdot e^{3\sqrt{l}}$.

□